

MAT101R01 – ENGINEERING MATHEMATICS – I

(Common for all branches of Engineering) Semester I

Unit I: Trigonometry

Expansions of $\sin n\theta$ and $\cos n\theta$ in powers of $\sin \theta$ and $\cos \theta$ – Expansions of $\sin^n \theta$ and $\cos^n \theta$ in terms of sines and cosines of multiples of θ – Hyperbolic and inverse hyperbolic functions – Logarithm of complex numbers – Separation of complex functions into real and imaginary parts.

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1 Expansions of $\cos n\theta$, $\sin n\theta$ and $\tan n\theta$

1.1 Tools required

Result 1.1 (De Moivre's Theorem)

For any integer n ,

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta.$$

Result 1.2 (Binomial expansion)

$$(x + a)^n = x^n + {}^nC_1 x^{n-1}a + {}^nC_2 x^{n-2}a^2 + {}^nC_3 x^{n-3}a^3 + \cdots + {}^nC_n a^n, \quad {}^nC_r = \frac{n!}{(n-r)!r!}, \quad {}^nC_r = {}^nC_{n-r}.$$

For example,

$${}^5C_2 = \frac{5!}{3!2!} = \frac{5 \times 4}{2} = 10, \quad {}^5C_3 = \frac{5!}{2!3!} = 10, \quad {}^5C_4 = {}^5C_1 = 5.$$

The binomial coefficients may also be read off from **Pascal's triangle**:

$$\begin{array}{rcccccccc} n=1: & & & & 1 & & 1 & & \\ n=2: & & & 1 & 2 & 1 & & & \\ n=3: & & 1 & 3 & 3 & 1 & & & \\ n=4: & & & 1 & 4 & 6 & 4 & 1 & \\ n=5: & & 1 & 5 & 10 & 10 & 5 & 1 & \\ n=6: & 1 & 6 & 15 & 20 & 15 & 6 & 1 & \\ n=7: & 1 & 7 & 21 & 35 & 35 & 21 & 7 & 1 \end{array}$$

We shall also repeatedly use $i^2 = -1$, $i^3 = -i$, $i^4 = 1$.

1.2 Expansion of $\cos 5\theta$ and $\sin 5\theta$

Problem 1.1

Expand $\cos 5\theta$ and $\sin 5\theta$ in powers of $\cos \theta$ and $\sin \theta$.

Solution. By De Moivre's theorem,

$$\cos 5\theta + i \sin 5\theta = (\cos \theta + i \sin \theta)^5.$$

By the binomial expansion,

$$\begin{aligned} \cos 5\theta + i \sin 5\theta &= \cos^5 \theta + {}^5C_1 \cos^4 \theta (i \sin \theta) + {}^5C_2 \cos^3 \theta (i \sin \theta)^2 + {}^5C_3 \cos^2 \theta (i \sin \theta)^3 \\ &\quad + {}^5C_4 \cos \theta (i \sin \theta)^4 + {}^5C_5 (i \sin \theta)^5 \\ &= \cos^5 \theta + 5 \cos^4 \theta (i \sin \theta) + 10 \cos^3 \theta (-\sin^2 \theta) + 10 \cos^2 \theta (-i \sin^3 \theta) \\ &\quad + 5 \cos \theta (\sin^4 \theta) + i \sin^5 \theta \\ &= \cos^5 \theta + i 5 \cos^4 \theta \sin \theta - 10 \cos^3 \theta \sin^2 \theta - i 10 \cos^2 \theta \sin^3 \theta + 5 \cos \theta \sin^4 \theta + i \sin^5 \theta. \end{aligned}$$

Equating the real parts, we get

$$\cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta$$

If we equate the imaginary parts, we get the expansion for $\sin 5\theta$:

$$\sin 5\theta = 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta$$

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Note

In the above expansion of $\cos 5\theta$, if we replace $\sin^2 \theta = 1 - \cos^2 \theta$, we get $\cos 5\theta$ in terms of powers of $\cos \theta$ only:

$$\begin{aligned}\cos 5\theta &= \cos^5 \theta - 10 \cos^3 \theta (1 - \cos^2 \theta) + 5 \cos \theta (1 - \cos^2 \theta)^2 \\ &= \cos^5 \theta - 10 \cos^3 \theta + 10 \cos^5 \theta + 5 \cos \theta (1 + \cos^4 \theta - 2 \cos^2 \theta) \\ &= \cos^5 \theta - 10 \cos^3 \theta + 10 \cos^5 \theta + 5 \cos \theta + 5 \cos^5 \theta - 10 \cos^3 \theta \\ \Rightarrow \cos 5\theta &= 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta.\end{aligned}$$

Similarly, in $\sin 5\theta = 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta$, if we replace $\cos^2 \theta = 1 - \sin^2 \theta$, we get

$$\begin{aligned}\sin 5\theta &= 5(1 - \sin^2 \theta)^2 \sin \theta - 10(1 - \sin^2 \theta) \sin^3 \theta + \sin^5 \theta \\ &= 5 \sin \theta - 10 \sin^3 \theta + 5 \sin^5 \theta - 10 \sin^3 \theta + 10 \sin^5 \theta + \sin^5 \theta \\ \Rightarrow \sin 5\theta &= 16 \sin^5 \theta - 20 \sin^3 \theta + 5 \sin \theta.\end{aligned}$$

Result 1.3

$$\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta, \quad \sin 5\theta = 16 \sin^5 \theta - 20 \sin^3 \theta + 5 \sin \theta.$$

1.3 Expansion of $\sin 7\theta$ **Problem 1.2**

Obtain the expansion of $\sin 7\theta$ (in powers of $\sin \theta$).

Solution. By De Moivre's rule,

$$\begin{aligned}\cos 7\theta + i \sin 7\theta &= (\cos \theta + i \sin \theta)^7 \\ &= \cos^7 \theta + {}^7C_1 \cos^6 \theta (i \sin \theta) + {}^7C_2 \cos^5 \theta (i \sin \theta)^2 + {}^7C_3 \cos^4 \theta (i \sin \theta)^3 \\ &\quad + {}^7C_4 \cos^3 \theta (i \sin \theta)^4 + {}^7C_5 \cos^2 \theta (i \sin \theta)^5 + {}^7C_6 \cos \theta (i \sin \theta)^6 + {}^7C_7 (i \sin \theta)^7.\end{aligned}$$

Using $i^2 = -1$, $i^4 = 1$ and ${}^7C_1 = 7$, ${}^7C_3 = 35$, ${}^7C_5 = 21$, ${}^7C_7 = 1$, and equating imaginary parts,

$$\sin 7\theta = 7 \cos^6 \theta \sin \theta - 35 \cos^4 \theta \sin^3 \theta + 21 \cos^2 \theta \sin^5 \theta - \sin^7 \theta.$$

Now put

$$\cos^2 \theta = 1 - \sin^2 \theta, \quad \cos^4 \theta = (1 - \sin^2 \theta)^2, \quad \cos^6 \theta = (1 - \sin^2 \theta)^3,$$

where $(a - b)^3 = a^3 - b^3 - 3a^2b + 3ab^2$, i.e. $(1 - \sin^2 \theta)^3 = 1 - 3 \sin^2 \theta + 3 \sin^4 \theta - \sin^6 \theta$. Then

$$\begin{aligned}\sin 7\theta &= 7 \sin \theta (1 - 3 \sin^2 \theta + 3 \sin^4 \theta - \sin^6 \theta) - 35 \sin^3 \theta (1 - 2 \sin^2 \theta + \sin^4 \theta) \\ &\quad + 21 \sin^5 \theta (1 - \sin^2 \theta) - \sin^7 \theta \\ &= 7 \sin \theta - 21 \sin^3 \theta + 21 \sin^5 \theta - 7 \sin^7 \theta - 35 \sin^3 \theta + 70 \sin^5 \theta - 35 \sin^7 \theta \\ &\quad + 21 \sin^5 \theta - 21 \sin^7 \theta - \sin^7 \theta.\end{aligned}$$

Hence

$$\boxed{\sin 7\theta = 7 \sin \theta - 56 \sin^3 \theta + 112 \sin^5 \theta - 64 \sin^7 \theta}$$

or

$$\boxed{\frac{\sin 7\theta}{\sin \theta} = 7 - 56 \sin^2 \theta + 112 \sin^4 \theta - 64 \sin^6 \theta}$$

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1.4 General formulae for $\cos n\theta$, $\sin n\theta$ and $\tan n\theta$

In the same way we can derive the general formulae for $\cos n\theta$ and $\sin n\theta$.

By De Moivre's rule,

$$\begin{aligned}
 \cos n\theta + i \sin n\theta &= (\cos \theta + i \sin \theta)^n \\
 &= \cos^n \theta + {}^nC_1 \cos^{n-1} \theta (i \sin \theta) + {}^nC_2 \cos^{n-2} \theta (i^2 \sin^2 \theta) \\
 &\quad + {}^nC_3 \cos^{n-3} \theta (i^3 \sin^3 \theta) + {}^nC_4 \cos^{n-4} \theta (i^4 \sin^4 \theta) + \dots \\
 &= \cos^n \theta + i {}^nC_1 \cos^{n-1} \theta \sin \theta - {}^nC_2 \cos^{n-2} \theta \sin^2 \theta \\
 &\quad - i {}^nC_3 \cos^{n-3} \theta \sin^3 \theta + {}^nC_4 \cos^{n-4} \theta \sin^4 \theta + \dots
 \end{aligned}$$

Equating real parts and imaginary parts, we get

Result 1.4 (Expansions of $\cos n\theta$ and $\sin n\theta$)

$$\begin{aligned}
 \cos n\theta &= \cos^n \theta - {}^nC_2 \cos^{n-2} \theta \sin^2 \theta + {}^nC_4 \cos^{n-4} \theta \sin^4 \theta - \dots \\
 \sin n\theta &= {}^nC_1 \cos^{n-1} \theta \sin \theta - {}^nC_3 \cos^{n-3} \theta \sin^3 \theta + {}^nC_5 \cos^{n-5} \theta \sin^5 \theta - \dots
 \end{aligned}$$

Hence

$$\tan n\theta = \frac{\sin n\theta}{\cos n\theta} = \frac{{}^nC_1 \cos^{n-1} \theta \sin \theta - {}^nC_3 \cos^{n-3} \theta \sin^3 \theta + \dots}{\cos^n \theta - {}^nC_2 \cos^{n-2} \theta \sin^2 \theta + {}^nC_4 \cos^{n-4} \theta \sin^4 \theta - \dots}.$$

Dividing both numerator and denominator by $\cos^n \theta$, we get

Result 1.5 (Expansion of $\tan n\theta$)

$$\tan n\theta = \frac{{}^nC_1 \tan \theta - {}^nC_3 \tan^3 \theta + {}^nC_5 \tan^5 \theta - \dots}{1 - {}^nC_2 \tan^2 \theta + {}^nC_4 \tan^4 \theta - \dots}$$

Exercises. Using the above method, expand (1) $\cos 8\theta$ (2) $\sin 6\theta$.

1.5 Problems

Problem 1.3

If $x = 2 \cos \theta$, prove that

$$\frac{1 + \cos 7\theta}{1 + \cos \theta} = (x^3 - x^2 - 2x + 1)^2.$$

Solution. We know that $\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$. Replacing θ by $\theta/2$,

$$\cos^2 \frac{\theta}{2} = \frac{1 + \cos \theta}{2} \Rightarrow \boxed{2 \cos^2 \frac{\theta}{2} = 1 + \cos \theta}$$

Therefore

$$\frac{1 + \cos 7\theta}{1 + \cos \theta} = \frac{2 \cos^2 \frac{7\theta}{2}}{2 \cos^2 \frac{\theta}{2}} = \left(\frac{\cos \frac{7\theta}{2}}{\cos \frac{\theta}{2}} \right)^2 = \left(\frac{\cos 7\alpha}{\cos \alpha} \right)^2, \quad \text{where } \alpha = \frac{\theta}{2}. \quad (1)$$

We know that

$$\cos n\theta = \cos^n \theta - {}^nC_2 \cos^{n-2} \theta \sin^2 \theta + {}^nC_4 \cos^{n-4} \theta \sin^4 \theta - \dots$$

so that (with ${}^7C_2 = 21$, ${}^7C_4 = 35$, ${}^7C_6 = 7$)

$$\begin{aligned}
 \cos 7\alpha &= \cos^7 \alpha - {}^7C_2 \cos^5 \alpha \sin^2 \alpha + {}^7C_4 \cos^3 \alpha \sin^4 \alpha - {}^7C_6 \cos \alpha \sin^6 \alpha \\
 &= \cos^7 \alpha - 21 \cos^5 \alpha \sin^2 \alpha + 35 \cos^3 \alpha \sin^4 \alpha - 7 \cos \alpha \sin^6 \alpha \\
 \Rightarrow \frac{\cos 7\alpha}{\cos \alpha} &= \cos^6 \alpha - 21 \cos^4 \alpha \sin^2 \alpha + 35 \cos^2 \alpha \sin^4 \alpha - 7 \sin^6 \alpha.
 \end{aligned}$$

Replace $\sin^2 \alpha = 1 - \cos^2 \alpha$:

$$\begin{aligned}
 \frac{\cos 7\alpha}{\cos \alpha} &= \cos^6 \alpha - 21 \cos^4 \alpha (1 - \cos^2 \alpha) + 35 \cos^2 \alpha (1 - \cos^2 \alpha)^2 - 7(1 - \cos^2 \alpha)^3 \\
 &= \cos^6 \alpha - 21 \cos^4 \alpha + 21 \cos^6 \alpha + 35 \cos^2 \alpha (1 + \cos^4 \alpha - 2 \cos^2 \alpha) \\
 &\quad - 7(1 - \cos^6 \alpha + 3 \cos^4 \alpha - 3 \cos^2 \alpha) \\
 &= \cos^6 \alpha - 21 \cos^4 \alpha + 21 \cos^6 \alpha + 35 \cos^2 \alpha + 35 \cos^6 \alpha - 70 \cos^4 \alpha \\
 &\quad - 7 + 7 \cos^6 \alpha - 21 \cos^4 \alpha + 21 \cos^2 \alpha \\
 \Rightarrow \frac{\cos 7\alpha}{\cos \alpha} &= 64 \cos^6 \alpha - 112 \cos^4 \alpha + 56 \cos^2 \alpha - 7.
 \end{aligned}$$

Hence (1) becomes

$$\frac{1 + \cos 7\theta}{1 + \cos \theta} = \left(\frac{\cos 7\alpha}{\cos \alpha} \right)^2 = (64 \cos^6 \alpha - 112 \cos^4 \alpha + 56 \cos^2 \alpha - 7)^2. \quad (2)$$

But it is given that (i) $x = 2 \cos \theta$, and also we have chosen (ii) $\alpha = \theta/2$. We know

$$\cos^2 \frac{\theta}{2} = \frac{1 + \cos \theta}{2} \Rightarrow \cos^2 \alpha = \frac{1 + \frac{x}{2}}{2} \quad (\text{by (i) and (ii)}) \Rightarrow \boxed{\cos^2 \alpha = \frac{x+2}{4}}$$

Substituting this in (2),

$$\begin{aligned}
 \frac{1 + \cos 7\theta}{1 + \cos \theta} &= \left[64 \left(\frac{x+2}{4} \right)^3 - 112 \left(\frac{x+2}{4} \right)^2 + 56 \left(\frac{x+2}{4} \right) - 7 \right]^2 \\
 &= [(x+2)^3 - 7(x+2)^2 + 14(x+2) - 7]^2 \\
 &= [x^3 + 8 + 12x + 6x^2 - 7(x^2 + 4 + 4x) + 14x + 28 - 7]^2 \\
 &= [x^3 + x^2(6-7) + x(12-28+14) + (8-28+28-7)]^2 \\
 \Rightarrow \frac{1 + \cos 7\theta}{1 + \cos \theta} &= (x^3 - x^2 - 2x + 1)^2, \quad \text{where } x = 2 \cos \theta.
 \end{aligned}$$

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Problem 1.4

If $x = 2 \cos \theta$, identify $\frac{1 + \cos 5\theta}{1 + \cos \theta}$ in terms of x .

Solution. Since $1 + \cos \theta = 2 \cos^2 \frac{\theta}{2}$,

$$\frac{1 + \cos 5\theta}{1 + \cos \theta} = \frac{2 \cos^2 \frac{5\theta}{2}}{2 \cos^2 \frac{\theta}{2}} = \left(\frac{\cos 5\alpha}{\cos \alpha} \right)^2, \quad \alpha = \frac{\theta}{2}.$$

From $\cos 5\alpha = 16 \cos^5 \alpha - 20 \cos^3 \alpha + 5 \cos \alpha$,

$$\frac{\cos 5\alpha}{\cos \alpha} = 16 \cos^4 \alpha - 20 \cos^2 \alpha + 5.$$

As in the previous problem, $\cos^2 \alpha = \frac{x+2}{4}$. Hence

$$\begin{aligned}
 \frac{1 + \cos 5\theta}{1 + \cos \theta} &= \left[16 \left(\frac{x+2}{4} \right)^2 - 20 \left(\frac{x+2}{4} \right) + 5 \right]^2 \\
 &= [(x+2)^2 - 5(x+2) + 5]^2 \\
 &= [x^2 + 4x + 4 - 5x - 10 + 5]^2 = (x^2 - x - 1)^2, \quad \text{where } x = 2 \cos \theta.
 \end{aligned}$$

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2 Expansions of $\cos^n \theta$ and $\sin^n \theta$ in terms of multiple angles

2.1 The basic relations

Let $x = \cos \theta + i \sin \theta$. Then $\frac{1}{x} = \cos \theta - i \sin \theta$, and by De Moivre's rule

$$x^n = \cos n\theta + i \sin n\theta, \quad \frac{1}{x^n} = \cos n\theta - i \sin n\theta.$$

Adding and subtracting,

Result 2.1

$$x + \frac{1}{x} = 2 \cos \theta, \quad x - \frac{1}{x} = 2i \sin \theta, \quad x^n + \frac{1}{x^n} = 2 \cos n\theta, \quad x^n - \frac{1}{x^n} = 2i \sin n\theta.$$

2.2 General formula for $\cos^n \theta$

Now $x + \frac{1}{x} = 2 \cos \theta \Rightarrow \left(x + \frac{1}{x}\right)^n = (2 \cos \theta)^n = 2^n \cos^n \theta$. Hence

$$\begin{aligned} 2^n \cos^n \theta &= \left(x + \frac{1}{x}\right)^n \\ &= x^n + {}^nC_1 x^{n-1} \frac{1}{x} + {}^nC_2 x^{n-2} \frac{1}{x^2} + \cdots + {}^nC_{n-2} x^2 \frac{1}{x^{n-2}} + {}^nC_{n-1} x \frac{1}{x^{n-1}} + {}^nC_n \frac{1}{x^n} \\ &= x^n + {}^nC_1 x^{n-2} + {}^nC_2 x^{n-4} + \cdots + {}^nC_2 \frac{1}{x^{n-4}} + {}^nC_1 \frac{1}{x^{n-2}} + \frac{1}{x^n} \\ &= \left(x^n + \frac{1}{x^n}\right) + {}^nC_1 \left(x^{n-2} + \frac{1}{x^{n-2}}\right) + {}^nC_2 \left(x^{n-4} + \frac{1}{x^{n-4}}\right) + \cdots \end{aligned}$$

(grouping terms equidistant from the ends, since ${}^nC_r = {}^nC_{n-r}$). We know $x^n + \frac{1}{x^n} = 2 \cos n\theta$, so

$$2^n \cos^n \theta = 2 \cos n\theta + {}^nC_1 (2 \cos(n-2)\theta) + {}^nC_2 (2 \cos(n-4)\theta) + \cdots$$

Result 2.2 (Expansion of $\cos^n \theta$)

$$\cos^n \theta = \frac{1}{2^{n-1}} \left[\cos n\theta + {}^nC_1 \cos(n-2)\theta + {}^nC_2 \cos(n-4)\theta + \cdots \right]$$

Similarly, from $\left(x - \frac{1}{x}\right)^n = (2i \sin \theta)^n$ we can expand $\sin^n \theta$, using

$$\left(x - \frac{1}{x}\right)^n = x^n - {}^nC_1 x^{n-1} \frac{1}{x} + {}^nC_2 x^{n-2} \frac{1}{x^2} - \cdots$$

2.3 Problems

Problem 2.1

Prove that $2^4 \cos^5 \theta = \cos 5\theta + 5 \cos 3\theta + 10 \cos \theta$.

Solution. Let $x = \cos \theta + i \sin \theta \Rightarrow \frac{1}{x} = \cos \theta - i \sin \theta$; also $x^n = \cos n\theta + i \sin n\theta$ and $\frac{1}{x^n} = \cos n\theta - i \sin n\theta$. Now $x + \frac{1}{x} = 2 \cos \theta$ and $x^n + \frac{1}{x^n} = 2 \cos n\theta$.

$$\begin{aligned} \Rightarrow (2 \cos \theta)^5 &= \left(x + \frac{1}{x}\right)^5 \\ \Rightarrow 2^5 \cos^5 \theta &= x^5 + {}^5C_1 x^4 \frac{1}{x} + {}^5C_2 x^3 \frac{1}{x^2} + {}^5C_3 x^2 \frac{1}{x^3} + {}^5C_4 x \frac{1}{x^4} + {}^5C_5 \frac{1}{x^5} \\ &= x^5 + 5x^3 + 10x + 10 \frac{1}{x} + 5 \frac{1}{x^3} + \frac{1}{x^5} \end{aligned}$$

$$\begin{aligned}
&= \left(x^5 + \frac{1}{x^5}\right) + 5 \left(x^3 + \frac{1}{x^3}\right) + 10 \left(x + \frac{1}{x}\right) \\
&= 2 \cos 5\theta + 5(2 \cos 3\theta) + 10(2 \cos \theta) \quad (\because x^n + \frac{1}{x^n} = 2 \cos n\theta) \\
\Rightarrow 2^4 \cos^5 \theta &= \cos 5\theta + 5 \cos 3\theta + 10 \cos \theta.
\end{aligned}$$

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Problem 2.2Expand $\sin^8 \theta$ in terms of $\cos n\theta$ (i.e. in a series of cosines of multiples of θ).

Solution. Let $x = \cos \theta + i \sin \theta$, $\frac{1}{x} = \cos \theta - i \sin \theta$, $x^n = \cos n\theta + i \sin n\theta$, $\frac{1}{x^n} = \cos n\theta - i \sin n\theta$. Then $x - \frac{1}{x} = 2i \sin \theta$.

$$\begin{aligned}
\Rightarrow (2i \sin \theta)^8 &= \left(x - \frac{1}{x}\right)^8 \\
\Rightarrow 2^8 i^8 \sin^8 \theta &= x^8 - {}^8C_1 x^7 \frac{1}{x} + {}^8C_2 x^6 \frac{1}{x^2} - {}^8C_3 x^5 \frac{1}{x^3} + {}^8C_4 x^4 \frac{1}{x^4} - {}^8C_5 x^3 \frac{1}{x^5} \\
&\quad + {}^8C_6 x^2 \frac{1}{x^6} - {}^8C_7 x \frac{1}{x^7} + {}^8C_8 \frac{1}{x^8} \\
&= x^8 - 8x^6 + 28x^4 - 56x^2 + 70 - 56 \frac{1}{x^2} + 28 \frac{1}{x^4} - 8 \frac{1}{x^6} + \frac{1}{x^8}.
\end{aligned}$$

Since $i^8 = (i^4)^2 = 1$,

$$\begin{aligned}
2^8 \sin^8 \theta &= \left(x^8 + \frac{1}{x^8}\right) - 8 \left(x^6 + \frac{1}{x^6}\right) + 28 \left(x^4 + \frac{1}{x^4}\right) - 56 \left(x^2 + \frac{1}{x^2}\right) + 70 \\
&= 2 \cos 8\theta - 8(2 \cos 6\theta) + 28(2 \cos 4\theta) - 56(2 \cos 2\theta) + 70 \quad (\because x^n + \frac{1}{x^n} = 2 \cos n\theta) \\
\Rightarrow \sin^8 \theta &= \frac{1}{2^7} [\cos 8\theta - 8 \cos 6\theta + 28 \cos 4\theta - 56 \cos 2\theta + 35].
\end{aligned}$$

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Problem 2.3Expand $\sin^7 \theta$ in a series of sines of multiples of θ .

Solution. With $x = \cos \theta + i \sin \theta$ we have $x - \frac{1}{x} = 2i \sin \theta$ and $x^n - \frac{1}{x^n} = 2i \sin n\theta$. Also $i^7 = i^6 \cdot i = (i^4)(i^2) i = -i$.

$$\begin{aligned}
(2i \sin \theta)^7 &= \left(x - \frac{1}{x}\right)^7 \\
\Rightarrow 2^7 i^7 \sin^7 \theta &= x^7 - 7x^5 + 21x^3 - 35x + 35 \frac{1}{x} - 21 \frac{1}{x^3} + 7 \frac{1}{x^5} - \frac{1}{x^7} \\
\Rightarrow -2^7 i \sin^7 \theta &= \left(x^7 - \frac{1}{x^7}\right) - 7 \left(x^5 - \frac{1}{x^5}\right) + 21 \left(x^3 - \frac{1}{x^3}\right) - 35 \left(x - \frac{1}{x}\right) \\
&= 2i \sin 7\theta - 7(2i \sin 5\theta) + 21(2i \sin 3\theta) - 35(2i \sin \theta).
\end{aligned}$$

Dividing by $-2^7 i$,

$$\sin^7 \theta = -\frac{1}{64} [\sin 7\theta - 7 \sin 5\theta + 21 \sin 3\theta - 35 \sin \theta]$$

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Problem 2.4Expand $\cos^7 \theta \sin^6 \theta$ in a series of cosines of multiples of θ .

Solution. Let $x = \cos \theta + i \sin \theta$, $\frac{1}{x} = \cos \theta - i \sin \theta$, so that $x + \frac{1}{x} = 2 \cos \theta$ and $x - \frac{1}{x} = 2i \sin \theta$.

$$\Rightarrow (\cos \theta)^7 (\sin \theta)^6 = \left(x + \frac{1}{x}\right)^7 \left(x - \frac{1}{x}\right)^6$$

$$\begin{aligned}
\Rightarrow 2^{13} i^6 \cos^7 \theta \sin^6 \theta &= \left(x + \frac{1}{x}\right) \left(x + \frac{1}{x}\right)^6 \left(x - \frac{1}{x}\right)^6 \\
&= \left(x + \frac{1}{x}\right) \left[\left(x + \frac{1}{x}\right) \left(x - \frac{1}{x}\right)\right]^6 \\
&= \left(x + \frac{1}{x}\right) \left(x^2 - \frac{1}{x^2}\right)^6 \quad (\because (a+b)(a-b) = a^2 - b^2) \\
&= \left(x + \frac{1}{x}\right) \left[(x^2)^6 - {}^6C_1(x^2)^5 \frac{1}{x^2} + {}^6C_2(x^2)^4 \left(\frac{1}{x^2}\right)^2 - {}^6C_3(x^2)^3 \left(\frac{1}{x^2}\right)^3 \right. \\
&\quad \left. + {}^6C_4(x^2)^2 \left(\frac{1}{x^2}\right)^4 - {}^6C_5(x^2) \left(\frac{1}{x^2}\right)^5 + {}^6C_6 \left(\frac{1}{x^2}\right)^6\right] \\
&= \left(x + \frac{1}{x}\right) \left[x^{12} - 6x^8 + 15x^4 - 20 + 15\frac{1}{x^4} - 6\frac{1}{x^8} + \frac{1}{x^{12}}\right] \\
&= x^{13} - 6x^9 + 15x^5 - 20x + \frac{15}{x^3} - \frac{6}{x^7} + \frac{1}{x^{11}} \\
&\quad + x^{11} - 6x^7 + 15x^3 - \frac{20}{x} + \frac{15}{x^5} - \frac{6}{x^9} + \frac{1}{x^{13}} \\
&= \left(x^{13} + \frac{1}{x^{13}}\right) + \left(x^{11} + \frac{1}{x^{11}}\right) - 6\left(x^9 + \frac{1}{x^9}\right) - 6\left(x^7 + \frac{1}{x^7}\right) \\
&\quad + 15\left(x^5 + \frac{1}{x^5}\right) + 15\left(x^3 + \frac{1}{x^3}\right) - 20\left(x + \frac{1}{x}\right) \\
&= 2 \cos 13\theta + 2 \cos 11\theta - 6(2 \cos 9\theta) - 6(2 \cos 7\theta) + 15(2 \cos 5\theta) + 15(2 \cos 3\theta) - 20(2 \cos \theta).
\end{aligned}$$

Since $i^6 = -1$, dividing by -2^{13} :

$$\cos^7 \theta \sin^6 \theta = -\frac{1}{2^{12}} \left[\cos 13\theta + \cos 11\theta - 6 \cos 9\theta - 6 \cos 7\theta + 15 \cos 5\theta + 15 \cos 3\theta - 20 \cos \theta \right]$$

■

Problem 2.5

Expand $\cos^6 \theta \sin^5 \theta$ in a series of sines of multiples of θ .

Solution. With $x + \frac{1}{x} = 2 \cos \theta$, $x - \frac{1}{x} = 2i \sin \theta$:

$$\begin{aligned}
(2 \cos \theta)^6 (2i \sin \theta)^5 &= \left(x + \frac{1}{x}\right)^6 \left(x - \frac{1}{x}\right)^5 \\
\Rightarrow 2^{11} i \cos^6 \theta \sin^5 \theta &= \left(x + \frac{1}{x}\right) \left(x + \frac{1}{x}\right)^5 \left(x - \frac{1}{x}\right)^5 \quad (i^5 = i) \\
&= \left(x + \frac{1}{x}\right) \left[\left(x + \frac{1}{x}\right) \left(x - \frac{1}{x}\right)\right]^5 = \left(x + \frac{1}{x}\right) \left(x^2 - \frac{1}{x^2}\right)^5 \\
&= \left(x + \frac{1}{x}\right) \left[(x^2)^5 - {}^5C_1(x^2)^4 \frac{1}{x^2} + {}^5C_2(x^2)^3 \left(\frac{1}{x^2}\right)^2 - {}^5C_3(x^2)^2 \left(\frac{1}{x^2}\right)^3 \right. \\
&\quad \left. + {}^5C_4(x^2) \left(\frac{1}{x^2}\right)^4 - {}^5C_5 \left(\frac{1}{x^2}\right)^5\right] \\
&= \left(x + \frac{1}{x}\right) \left[x^{10} - 5x^6 + 10x^2 - 10\frac{1}{x^2} + 5\frac{1}{x^6} - \frac{1}{x^{10}}\right] \\
&= x^{11} - 5x^7 + 10x^3 - \frac{10}{x} + \frac{5}{x^5} - \frac{1}{x^9} + x^9 - 5x^5 + 10x - \frac{10}{x^3} + \frac{5}{x^7} - \frac{1}{x^{11}} \\
&= \left(x^{11} - \frac{1}{x^{11}}\right) + \left(x^9 - \frac{1}{x^9}\right) - 5\left(x^7 - \frac{1}{x^7}\right) - 5\left(x^5 - \frac{1}{x^5}\right) \\
&\quad + 10\left(x^3 - \frac{1}{x^3}\right) + 10\left(x - \frac{1}{x}\right).
\end{aligned}$$

Using $x^n - \frac{1}{x^n} = 2i \sin n\theta$,

$$2^{11} i \cos^6 \theta \sin^5 \theta = 2i \sin 11\theta + 2i \sin 9\theta - 5(2i \sin 7\theta) - 5(2i \sin 5\theta) + 10(2i \sin 3\theta) + 10(2i \sin \theta)$$

$$\Rightarrow \cos^6 \theta \sin^5 \theta = \frac{1}{2^{10}} \left[\sin 11\theta + \sin 9\theta - 5 \sin 7\theta - 5 \sin 5\theta + 10 \sin 3\theta + 10 \sin \theta \right]$$

■

Problem 2.6Expand $\cos^5 \theta \sin^4 \theta$ in a series of cosines of multiples of θ .**Solution.** Proceeding as above, $(2 \cos \theta)^5 (2i \sin \theta)^4 = (x + \frac{1}{x}) (x^2 - \frac{1}{x^2})^4$, and $i^4 = 1$:

$$\begin{aligned} 2^9 \cos^5 \theta \sin^4 \theta &= \left(x + \frac{1}{x}\right) \left[x^8 - 4x^4 + 6 - \frac{4}{x^4} + \frac{1}{x^8}\right] \\ &= \left(x^9 + \frac{1}{x^9}\right) + \left(x^7 + \frac{1}{x^7}\right) - 4 \left(x^5 + \frac{1}{x^5}\right) - 4 \left(x^3 + \frac{1}{x^3}\right) + 6 \left(x + \frac{1}{x}\right) \\ &= 2 \cos 9\theta + 2 \cos 7\theta - 8 \cos 5\theta - 8 \cos 3\theta + 12 \cos \theta \\ \Rightarrow \cos^5 \theta \sin^4 \theta &= \frac{1}{2^8} \left[\cos 9\theta + \cos 7\theta - 4 \cos 5\theta - 4 \cos 3\theta + 6 \cos \theta \right]. \end{aligned}$$

■

Problem 2.7Show that $16 \sin^5 \theta = \sin 5\theta - 5 \sin 3\theta + 10 \sin \theta$. Hence solve completely the equation

$$\cos 5\theta + 5 \cos 3\theta + 10 \cos \theta = \frac{1}{2}, \quad \text{where } \theta \text{ is real.}$$

Solution. Let $x = \cos \theta + i \sin \theta$; then $\frac{1}{x} = \cos \theta - i \sin \theta$, so $x - \frac{1}{x} = 2i \sin \theta$.

$$\begin{aligned} \Rightarrow (2i \sin \theta)^5 &= \left(x - \frac{1}{x}\right)^5 \\ 2^5 i^5 \sin^5 \theta &= x^5 - {}^5C_1 x^4 \frac{1}{x} + {}^5C_2 x^3 \frac{1}{x^2} - {}^5C_3 x^2 \frac{1}{x^3} + {}^5C_4 x \frac{1}{x^4} - {}^5C_5 \frac{1}{x^5} \\ 2^5 i \sin^5 \theta &= x^5 - 5x^3 + 10x - \frac{10}{x} + \frac{5}{x^3} - \frac{1}{x^5} \\ &= \left(x^5 - \frac{1}{x^5}\right) - 5 \left(x^3 - \frac{1}{x^3}\right) + 10 \left(x - \frac{1}{x}\right) \\ 2^5 i \sin^5 \theta &= (2i \sin 5\theta) - 5(2i \sin 3\theta) + 10(2i \sin \theta) \quad \left(\because x^n - \frac{1}{x^n} = 2i \sin n\theta\right) \\ \Rightarrow 2^4 \sin^5 \theta &= \sin 5\theta - 5 \sin 3\theta + 10 \sin \theta \\ \Rightarrow \boxed{16 \sin^5 \theta &= \sin 5\theta - 5 \sin 3\theta + 10 \sin \theta} \end{aligned} \tag{1}$$

Put $\theta = \frac{\pi}{2} - \theta$ in the above equation (1). Then

$$\begin{aligned} 16 \left(\sin \left(\frac{\pi}{2} - \theta\right)\right)^5 &= \sin 5 \left(\frac{\pi}{2} - \theta\right) - 5 \sin 3 \left(\frac{\pi}{2} - \theta\right) + 10 \sin \left(\frac{\pi}{2} - \theta\right) \\ 16(\cos \theta)^5 &= \sin \left(\frac{5\pi}{2} - 5\theta\right) - 5 \sin \left(\frac{3\pi}{2} - 3\theta\right) + 10 \cos \theta. \end{aligned}$$

Since $\sin \left(\frac{5\pi}{2} - 5\theta\right) = \cos 5\theta$ and $\sin \left(\frac{3\pi}{2} - 3\theta\right) = -\cos 3\theta$ (ASTC rule),

$$\boxed{16 \cos^5 \theta = \cos 5\theta + 5 \cos 3\theta + 10 \cos \theta} \tag{2}$$

But it is given that $\cos 5\theta + 5 \cos 3\theta + 10 \cos \theta = \frac{1}{2}$. Therefore by (2), we get

$$16 \cos^5 \theta = \frac{1}{2} \Rightarrow \cos^5 \theta = \frac{1}{32} = \frac{1}{2^5} \Rightarrow \cos \theta = \frac{1}{2} \quad (\theta \text{ real}) \Rightarrow \boxed{\theta = 2n\pi \pm \frac{\pi}{3}}, \quad n \in \mathbb{Z}.$$

■

Problem 2.8

Obtain the expansion of $\frac{\sin 7\theta}{\sin \theta}$ in a series of powers of $\cos 2\theta$ and hence find the equation whose roots are $\cos \frac{2\pi}{7}$, $\cos \frac{4\pi}{7}$, $\cos \frac{6\pi}{7}$.

Solution. Already we know that

$$\frac{\sin 7\theta}{\sin \theta} = 7 - 56 \sin^2 \theta + 112 \sin^4 \theta - 64 \sin^6 \theta.$$

Replace $\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$. Then

$$\begin{aligned} \frac{\sin 7\theta}{\sin \theta} &= 7 - 56 \left(\frac{1 - \cos 2\theta}{2} \right) + 112 \left(\frac{1 - \cos 2\theta}{2} \right)^2 - 64 \left(\frac{1 - \cos 2\theta}{2} \right)^3 \\ &= 7 - 28(1 - \cos 2\theta) + 28(1 + \cos^2 2\theta - 2 \cos 2\theta) - 8(1 - \cos^3 2\theta + 3 \cos^2 2\theta - 3 \cos 2\theta) \\ &= (7 - 28 + 28 - 8) + \cos 2\theta (28 - 56 + 24) + \cos^2 2\theta (28 - 24) + \cos^3 2\theta (8) \\ \Rightarrow \frac{\sin 7\theta}{\sin \theta} &= -1 - 4 \cos 2\theta + 4 \cos^2 2\theta + 8 \cos^3 2\theta. \end{aligned} \quad (1)$$

We know $\sin 7\theta = 0$ when $\theta = 0, \pm \frac{\pi}{7}, \pm \frac{2\pi}{7}, \pm \frac{3\pi}{7}, \pm \frac{4\pi}{7}, \dots$, but $\frac{\sin 7\theta}{\sin \theta} = 0$ only when $\theta = \pm \frac{\pi}{7}, \pm \frac{2\pi}{7}, \pm \frac{3\pi}{7}, \dots$ (not $\theta = 0$). In particular $\frac{\sin 7\theta}{\sin \theta} = 0$ when $\theta = \frac{\pi}{7}, \frac{2\pi}{7}, \frac{3\pi}{7}$. Therefore by (1),

$$8 \cos^3 2\theta + 4 \cos^2 2\theta - 4 \cos 2\theta - 1 = 0 \quad \text{for } \theta = \frac{\pi}{7}, \frac{2\pi}{7}, \frac{3\pi}{7}.$$

When $x = \cos 2\theta$, the values $\cos \frac{2\pi}{7}, \cos \frac{4\pi}{7}, \cos \frac{6\pi}{7}$ are the roots of $8x^3 + 4x^2 - 4x - 1 = 0$.

That is, $8x^3 + 4x^2 - 4x - 1 = 0$ (where $x = \cos 2\theta$) is the equation whose roots are $\cos \frac{2\pi}{7}, \cos \frac{4\pi}{7}$ and $\cos \frac{6\pi}{7}$. ■

Problem 2.9

Expand $\tan 7\theta$ in terms of powers of $\tan \theta$ and hence find the equation whose roots are $\tan \frac{k\pi}{7}, k = 0, 1, 2, \dots, 6$.

Solution. We know that

$$\begin{aligned} \tan n\theta &= \frac{{}^nC_1 \tan \theta - {}^nC_3 \tan^3 \theta + {}^nC_5 \tan^5 \theta - \dots}{1 - {}^nC_2 \tan^2 \theta + {}^nC_4 \tan^4 \theta - \dots} \\ \Rightarrow \tan 7\theta &= \frac{{}^7C_1 \tan \theta - {}^7C_3 \tan^3 \theta + {}^7C_5 \tan^5 \theta - {}^7C_7 \tan^7 \theta}{1 - {}^7C_2 \tan^2 \theta + {}^7C_4 \tan^4 \theta - {}^7C_6 \tan^6 \theta} = \frac{7 \tan \theta - 35 \tan^3 \theta + 21 \tan^5 \theta - \tan^7 \theta}{1 - 21 \tan^2 \theta + 35 \tan^4 \theta - 7 \tan^6 \theta}. \end{aligned} \quad (1)$$

We know that $\tan 7\theta = 0$ when $\theta = 0, \frac{\pi}{7}, \frac{2\pi}{7}, \frac{3\pi}{7}, \dots$. Therefore by (1),

$${}^7C_1 \tan \theta - {}^7C_3 \tan^3 \theta + {}^7C_5 \tan^5 \theta - {}^7C_7 \tan^7 \theta = 0 \quad \text{for } \theta = 0, \frac{\pi}{7}, \frac{2\pi}{7}, \frac{3\pi}{7}, \dots$$

$$\Rightarrow 7 \tan \theta - 35 \tan^3 \theta + 21 \tan^5 \theta - \tan^7 \theta = 0 \quad \text{for } \theta = 0, \frac{\pi}{7}, \frac{2\pi}{7}, \dots, \frac{6\pi}{7}.$$

Therefore, when $x = \tan \theta$, the values $x = \tan 0, \tan \frac{\pi}{7}, \tan \frac{2\pi}{7}, \dots, \tan \frac{6\pi}{7}$ are the roots of $7x - 35x^3 + 21x^5 - x^7 = 0$.

That is, $x^7 - 21x^5 + 35x^3 - 7x = 0$ is the required equation whose roots are $x = \tan \frac{k\pi}{7}, k = 0, 1, 2, \dots, 6$. ■

Exercises.

- (1) Expand $\frac{\cos 7\theta}{\cos^7 \theta}$ in a series of powers of $\tan \theta$.
- (2) Expand $\sin 6\theta \sec^6 \theta$ in a series of powers of $\tan \theta$.
- (3) Expand $\cot 5\theta$ in terms of powers of $\cot \theta$.
- (4) If $x = 2 \cos \theta$, prove that $2(1 + \cos 8\theta) = (x^4 - 4x^2 + 2)^2$.

3 Hyperbolic Functions

Certain even and odd combinations of the exponential functions e^x and e^{-x} (more precisely of $\frac{1}{2}e^x$ and $\frac{1}{2}e^{-x}$) occur so frequently in mathematics and its applications that they are given special names. They are similar to trigonometric functions: just as the trigonometric (circular) functions are related to the circle, these functions are related to the hyperbola. Hence they are called *hyperbolic functions*.

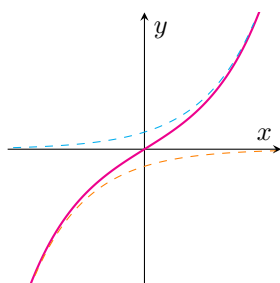
Definition 3.1 (Hyperbolic functions)

$$\cosh x = \frac{e^x + e^{-x}}{2} = \frac{1}{2}e^x + \frac{1}{2}e^{-x}, \quad \sinh x = \frac{e^x - e^{-x}}{2} = \frac{1}{2}e^x - \frac{1}{2}e^{-x},$$

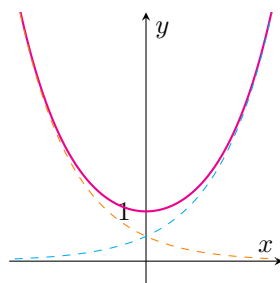
$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}},$$

$$\operatorname{cosech} x = \frac{1}{\sinh x}, \quad \operatorname{sech} x = \frac{1}{\cosh x}, \quad \operatorname{coth} x = \frac{\cosh x}{\sinh x}.$$

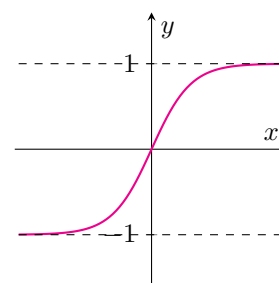
$y = \sinh x$



$y = \cosh x$



$y = \tanh x$

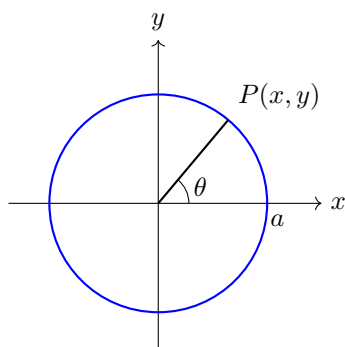


Note

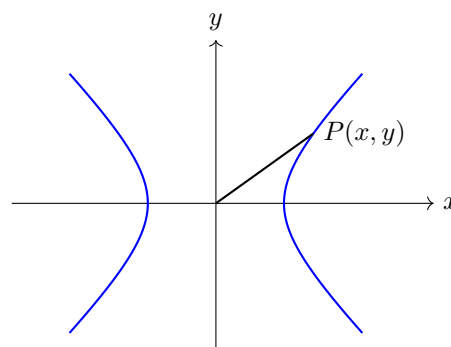
$\sinh x$ has domain $\mathbb{R}$ and range $\mathbb{R}$; $\cosh x$ has domain $\mathbb{R}$ and range $[1, \infty)$; $\tanh x$ has domain $\mathbb{R}$, range $(-1, 1)$ and horizontal asymptotes $y = \pm 1$. A well-known application: a heavy flexible cable (telephone or power line) suspended between two points at the same height hangs in the shape of the curve $y = c + a \cosh(x/a)$, called a *catenary*. Hyperbolic functions also appear in the description of ocean waves, e.g. the velocity of a water wave of length L over water of depth d is $v = \sqrt{\frac{gL}{2\pi}} \tanh \frac{2\pi d}{L}$.

Circular versus hyperbolic functions

Like the trigonometric functions having a relation with the circle, the hyperbolic functions have the same kind of relation with the hyperbola.



$$x^2 + y^2 = a^2: x = a \cos \theta, y = a \sin \theta$$



$$x^2 - y^2 = 1: x = \cosh \theta, y = \sinh \theta$$

We know the hyperbola equation $x^2 - y^2 = 1$; with $x = \cosh \theta$, $y = \sinh \theta$ we get $\cosh^2 \theta - \sinh^2 \theta = 1$ (for

the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$: $x = a \cosh \theta$, $y = b \sinh \theta$). In circular trigonometric functions we have the relation $\cos^2 \theta + \sin^2 \theta = 1$.

3.1 Hyperbolic identities

Result 3.1

- (1) $\cosh^2 x - \sinh^2 x = 1$
- (2) $\cosh^2 x + \sinh^2 x = \cosh 2x$
- (3) $\cosh^2 x = \frac{1 + \cosh 2x}{2}$
- (4) $\sinh^2 x = \frac{\cosh 2x - 1}{2}$

Proof of (1).

$$\begin{aligned} \text{LHS} = \cosh^2 x - \sinh^2 x &= \left(\frac{e^x + e^{-x}}{2} \right)^2 - \left(\frac{e^x - e^{-x}}{2} \right)^2 \\ &= \frac{1}{4} (e^{2x} + e^{-2x} + 2e^x e^{-x}) - \frac{1}{4} (e^{2x} + e^{-2x} - 2e^x e^{-x}) \\ &= \frac{1}{4} [e^{2x} + e^{-2x} + 2 - e^{2x} - e^{-2x} + 2] = 1 = \text{RHS}. \end{aligned}$$

Proof of (2).

$$\begin{aligned} \text{LHS} = \cosh^2 x + \sinh^2 x &= \left(\frac{e^x + e^{-x}}{2} \right)^2 + \left(\frac{e^x - e^{-x}}{2} \right)^2 \\ &= \frac{1}{4} (e^{2x} + e^{-2x} + 2) + \frac{1}{4} (e^{2x} + e^{-2x} - 2) \\ &= \frac{1}{4} [e^{2x} + e^{-2x} + 2 + e^{2x} + e^{-2x} - 2] = \frac{1}{4} [2(e^{2x} + e^{-2x})] \\ &= \frac{e^{2x} + e^{-2x}}{2} = \cosh 2x = \text{RHS} \quad \left(\because \cosh x = \frac{e^x + e^{-x}}{2} \right), \end{aligned}$$

whereas in trigonometry, $\cos^2 \theta - \sin^2 \theta = \cos 2\theta$.

Proof of (3). (Compare with $\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$.)

$$\begin{aligned} \text{LHS} = \cosh^2 x &= \left(\frac{e^x + e^{-x}}{2} \right)^2 = \frac{1}{4} (e^{2x} + e^{-2x} + 2) \\ &= \frac{1}{2} \left[\frac{e^{2x} + e^{-2x}}{2} + 1 \right] = \frac{1}{2} (1 + \cosh 2x) = \text{RHS}. \end{aligned}$$

Proof of (4). (Compare with $\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$.)

$$\begin{aligned} \text{LHS} = \sinh^2 x &= \left(\frac{e^x - e^{-x}}{2} \right)^2 = \frac{1}{4} (e^{2x} + e^{-2x} - 2) \\ &= \frac{1}{2} \left[\frac{e^{2x} + e^{-2x}}{2} - 1 \right] = \frac{1}{2} (\cosh 2x - 1) = \text{RHS}. \end{aligned}$$

Result 3.2 (Further hyperbolic identities)

$$\begin{aligned} \sinh(-x) &= -\sinh x, & \cosh(-x) &= \cosh x, & 1 - \tanh^2 x &= \text{sech}^2 x, \\ \sinh(x + y) &= \sinh x \cosh y + \cosh x \sinh y, & \cosh(x + y) &= \cosh x \cosh y + \sinh x \sinh y. \end{aligned}$$

For instance, dividing $\cosh^2 x - \sinh^2 x = 1$ by $\cosh^2 x$ gives $1 - \tanh^2 x = \text{sech}^2 x$.

3.2 Relation between trigonometric (circular) and hyperbolic functions

From the identities we see that there is some relationship between hyperbolic functions and circular trigonometric functions.

Result 3.3

- | | |
|----------------------------|------------------------------|
| (1) $\cos(ix) = \cosh x$ | i.e. $\cosh x = \cos(ix)$ |
| (2) $\sin(ix) = i \sinh x$ | i.e. $\sinh x = -i \sin(ix)$ |
| (3) $\tan(ix) = i \tanh x$ | i.e. $\tanh x = -i \tan(ix)$ |

Proof. (i) We know that $\cos \theta = \frac{1}{2} (e^{i\theta} + e^{-i\theta})$. Put $\theta = ix$:

$$\cos(ix) = \frac{1}{2} [e^{i(ix)} + e^{-i(ix)}] = \frac{1}{2} [e^{-x} + e^x] = \frac{1}{2} (e^x + e^{-x}) \Rightarrow \boxed{\cos(ix) = \cosh x}$$

(ii) We know that $\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$. Put $\theta = ix$:

$$\begin{aligned} \sin(ix) &= \frac{1}{2i} (e^{i(ix)} - e^{-i(ix)}) = \frac{1}{2i} (e^{-x} - e^x) = -\frac{1}{i} \left(\frac{e^x - e^{-x}}{2} \right) = -\frac{1}{i} \sinh x = -\frac{1}{i} \times \frac{i}{i} \sinh x = \frac{-i}{-1} \sinh x. \\ &\Rightarrow \boxed{\sin(ix) = i \sinh x} \end{aligned}$$

(iii) We know that

$$\begin{aligned} \tanh x &= \frac{\sinh x}{\cosh x} = \frac{\frac{1}{i} \sin(ix)}{\cos(ix)} = \frac{-i \sin(ix)}{\cos(ix)} = -i \tan(ix) \\ \Rightarrow \tan(ix) &= -\frac{1}{i} \tanh x = -(-i) \tanh x \Rightarrow \boxed{\tan(ix) = i \tanh x} \end{aligned}$$

3.3 Tutorial problems

Problem 3.1 (Tutorial)

- (1) Expand $\frac{\sin 8\theta}{\cos \theta}$ in a series of powers of $\sin \theta$.
Ans: $8 \sin \theta - 80 \sin^3 \theta + 192 \sin^5 \theta - 128 \sin^7 \theta$.
- (2) Find the equation whose roots are $\tan \frac{k\pi}{5}$, $k = 1, 2, 3, 4$ (first expand $\tan 5\theta$).
Ans: $x^4 - 10x^2 + 5 = 0$.
- (3) If $x = 2 \cos \theta$, show that $\frac{1 + \cos 9\theta}{1 + \cos \theta} = (x^4 - x^3 - 3x^2 + 2x + 1)^2$.
- (4) Expand $\sin^9 \theta$ in terms of $\sin n\theta$.
Ans: $\frac{1}{256} [\sin 9\theta - 9 \sin 7\theta + 36 \sin 5\theta - 84 \sin 3\theta + 126 \sin \theta]$.
- (5) Expand $\sin^6 \theta \cos^4 \theta$ in a series of cosines of multiples of θ .
Ans: $-\frac{1}{29} (\cos 10\theta - 2 \cos 8\theta - 3 \cos 6\theta + 8 \cos 4\theta + 2 \cos 2\theta - 6)$.

3.4 Some more on hyperbolic functions

Recall $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$, so that $e^1 = e = 2.718\dots$ (Euler's constant), and

$$e = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x.$$

(For $x = 1, 10, 100, 1000, \dots$ the values of $(1 + \frac{1}{x})^x$ approach $2.718\dots$)

Also recall the series

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots, \quad e^{-x} = 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots,$$

$$\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots, \quad \cos \theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots$$

and Euler's formulae $\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$, $\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$.

Note

Using the relations $\cos(ix) = \cosh x$ and $\sin(ix) = i \sinh x$, hyperbolic identities follow at once from circular ones. For example, put $\theta = ix$ in $\cos^2 \theta + \sin^2 \theta = 1$:

$$(\cos(ix))^2 + (\sin(ix))^2 = 1 \Rightarrow (\cosh x)^2 + (i \sinh x)^2 = 1 \Rightarrow \cosh^2 x - \sinh^2 x = 1.$$

Similarly $\cos^2 \theta - \sin^2 \theta = \cos 2\theta$ gives $\cosh^2 x + \sinh^2 x = \cosh 2x$, and $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$ gives $\sinh 3x = 3 \sinh x + 4 \sinh^3 x$.

3.5 Making hyperbolic identities from trigonometric identities

Result 3.4 (Osborn's Rule)

To obtain a hyperbolic identity from a trigonometric identity:

- (i) wherever θ is present, replace it by hx : e.g. $\theta \rightarrow hx$, $2\theta \rightarrow h2x$, $3\theta \rightarrow h3x$, and so on;
- (ii) wherever there is a product of two sine (or tan) terms, replace them by $\sinh$ (or $\tanh$) *with the sign changed*: e.g. $\sin^3 \theta \rightarrow -\sinh^3 x$ (in $-4 \sin^3 \theta$), $\tan^2 \theta \rightarrow -\tanh^2 x$;
- (iii) there is no change of sign in terms involving cosines.

Hence we have the following hyperbolic identities from the trigonometric identities:

(1) $\sin 2\theta = 2 \sin \theta \cos \theta$	$\implies$	$\sinh 2x = 2 \sinh x \cosh x$
(2) $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$	$\implies$	$\cosh 2x = \cosh^2 x + \sinh^2 x$
(3) $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$	$\implies$	$\sinh 3x = 3 \sinh x + 4 \sinh^3 x$
(4) $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$	$\implies$	$\cosh 3x = 4 \cosh^3 x - 3 \cosh x$
(5) $\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$	$\implies$	$\tanh 3x = \frac{3 \tanh x + \tanh^3 x}{1 + 3 \tanh^2 x}$
(6) $\sin 2\theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}$	$\implies$	$\sinh 2x = \frac{2 \tanh x}{1 - \tanh^2 x}$
(7) $\cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$	$\implies$	$\cosh 2x = \frac{1 + \tanh^2 x}{1 - \tanh^2 x}$
(8) $\sin(A + B) = \sin A \cos B + \cos A \sin B$	$\implies$	$\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y$
(9) $\cos(A + B) = \cos A \cos B - \sin A \sin B$	$\implies$	$\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y$

and so on.

3.6 Problems on hyperbolic functions

Problem 3.2

Show that $(\cosh x + \sinh x)^n = \cosh nx + \sinh nx$.

Solution. Since $\cos(ix) = \cosh x$ and $\sin(ix) = i \sinh x$, i.e. $\sinh x = \frac{1}{i} \sin(ix)$,

$$\begin{aligned} (\cosh x + \sinh x)^n &= \left(\cos(ix) + \frac{1}{i} \sin(ix) \right)^n = (\cos(ix) - i \sin(ix))^n \\ &= \cos n(ix) - i \sin n(ix) \quad (\because \text{by De Moivre's rule}) \\ &= \cos(inx) - i \sin(inx) = \cosh nx - i(i \sinh nx) \\ &= \cosh nx + \sinh nx. \end{aligned}$$

Hence $(\cosh x + \sinh x)^n = \cosh nx + \sinh nx$. ■

Problem 3.3

Prove that $\frac{1 + \tanh x}{1 - \tanh x} = \cosh 2x + \sinh 2x$.

Solution.

$$\begin{aligned} \frac{1 + \tanh x}{1 - \tanh x} &= \frac{1 + \frac{\sinh x}{\cosh x}}{1 - \frac{\sinh x}{\cosh x}} \\ &= \frac{(\cosh x + \sinh x)/\cosh x}{(\cosh x - \sinh x)/\cosh x} = \frac{\cosh x + \sinh x}{\cosh x - \sinh x} \\ &= \frac{\cosh x + \sinh x}{\cosh x - \sinh x} \times \frac{\cosh x + \sinh x}{\cosh x + \sinh x} \\ &= \frac{\cosh^2 x + 2 \sinh x \cosh x + \sinh^2 x}{\cosh^2 x - \sinh^2 x} \\ &= (\cosh^2 x + \sinh^2 x) + 2 \sinh x \cosh x \quad (\because \cosh^2 x - \sinh^2 x = 1) \\ \Rightarrow \frac{1 + \tanh x}{1 - \tanh x} &= \cosh 2x + \sinh 2x. \end{aligned}$$

(OR) Consider

$$\begin{aligned} \cosh 2x + \sinh 2x &= \frac{1 + \tanh^2 x}{1 - \tanh^2 x} + \frac{2 \tanh x}{1 - \tanh^2 x} \\ &= \frac{1 + \tanh^2 x + 2 \tanh x}{1 - \tanh^2 x} \\ &= \frac{(1 + \tanh x)^2}{(1 - \tanh x)(1 + \tanh x)} = \frac{1 + \tanh x}{1 - \tanh x}. \end{aligned}$$
■

Problem 3.4

Prove that $(1 + \cosh x + \sinh x)^n = 2^n \cosh^n \left(\frac{x}{2} \right) \left[\cosh \left(\frac{nx}{2} \right) + \sinh \left(\frac{nx}{2} \right) \right]$.

Solution. Using $1 + \cosh x = 2 \cosh^2 \frac{x}{2}$ and $\sinh x = 2 \sinh \frac{x}{2} \cosh \frac{x}{2}$ (hyperbolic analogues of $1 + \cos \theta = 2 \cos^2 \frac{\theta}{2}$ and $\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$),

$$\begin{aligned} (1 + \cosh x + \sinh x)^n &= \left(2 \cosh^2 \frac{x}{2} + 2 \sinh \frac{x}{2} \cosh \frac{x}{2} \right)^n \\ &= \left(2 \cosh \frac{x}{2} \left(\cosh \frac{x}{2} + \sinh \frac{x}{2} \right) \right)^n \end{aligned}$$

$$\begin{aligned}
&= 2^n \cosh^n \frac{x}{2} \left(\cosh \frac{x}{2} + \sinh \frac{x}{2} \right)^n \\
&= 2^n \cosh^n \frac{x}{2} \left(\cosh \frac{nx}{2} + \sinh \frac{nx}{2} \right) \quad (\because \text{by Problem 1}).
\end{aligned}$$

■

Problem 3.5

If $\tanh\left(\frac{x}{2}\right) = \tan\left(\frac{x}{2}\right)$, prove that $\cos x \cosh x = 1$.

Solution. Given $\tanh \frac{x}{2} = \tan \frac{x}{2}$. We know

$$\cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}, \quad \cosh 2x = \frac{1 + \tanh^2 x}{1 - \tanh^2 x}.$$

Now

$$\begin{aligned}
\cos x \cosh x &= \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \times \frac{1 + \tanh^2 \frac{x}{2}}{1 - \tanh^2 \frac{x}{2}} \\
&= \frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} \times \frac{1 + \tan^2 \frac{x}{2}}{1 - \tan^2 \frac{x}{2}} \quad (\because \tanh \frac{x}{2} = \tan \frac{x}{2}) \\
\Rightarrow \cos x \cosh x &= 1.
\end{aligned}$$

■

Problem 3.6

If $\tanh x = \sin \theta$, then prove that $\sec \theta = \cosh x$.

Solution. Given $\tanh x = \sin \theta$. Now

$$\begin{aligned}
\sec \theta &= \frac{1}{\cos \theta} = \frac{1}{\sqrt{1 - \sin^2 \theta}} = \frac{1}{\sqrt{1 - \tanh^2 x}} \quad (\because \tanh x = \sin \theta) \\
&= \frac{1}{\sqrt{1 - \frac{\sinh^2 x}{\cosh^2 x}}} = \frac{\cosh x}{\sqrt{\cosh^2 x - \sinh^2 x}} = \frac{\cosh x}{\sqrt{1}} = \cosh x.
\end{aligned}$$

(Alternatively, $1 - \tanh^2 x = \operatorname{sech}^2 x$ directly gives $\sec^2 \theta = 1/(1 - \sin^2 \theta) = \cosh^2 x$.)

■

4 Separation into real and imaginary parts

Problem 4.1

Separate the real and imaginary parts of the following: (i) $\sin(x + iy)$ (ii) $\cot(x + iy)$ (iii) $\tan^{-1}(x + iy)$.

Solution. (i) Let $\sin(x + iy) = A + iB$. Then

$$\begin{aligned}
\Rightarrow \sin x \cos(iy) + \cos x \sin(iy) &= A + iB \\
\Rightarrow \sin x \cosh y + i \cos x \sinh y &= A + iB \quad (\because \cos(iy) = \cosh y, \sin(iy) = i \sinh y)
\end{aligned}$$

$\Rightarrow$ Real part = $A = \sin x \cosh y$; Imaginary part = $B = \cos x \sinh y$.

(ii) Using $2 \cos A \sin B = \sin(A + B) - \sin(A - B)$ and $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$,

$$\begin{aligned}
\cot(x + iy) &= \frac{\cos(x + iy)}{\sin(x + iy)} = \frac{2 \cos(x + iy)}{2 \sin(x + iy)} \times \frac{\sin(x - iy)}{\sin(x - iy)} \\
&= \frac{\sin[(x + iy) + (x - iy)] - \sin[(x + iy) - (x - iy)]}{\cos[(x + iy) - (x - iy)] - \cos[(x + iy) + (x - iy)]}
\end{aligned}$$

$$= \frac{\sin 2x - \sin(2iy)}{\cos(2iy) - \cos 2x} = \frac{\sin 2x - i \sinh 2y}{\cosh 2y - \cos 2x} \quad (\because \cos(ix) = \cosh x, \sin(ix) = i \sinh x).$$

$$\Rightarrow \text{Real part} = \frac{\sin 2x}{\cosh 2y - \cos 2x}, \quad \text{Imaginary part} = \frac{-\sinh 2y}{\cosh 2y - \cos 2x}.$$

(iii) Let $\tan^{-1}(A + iB) = x + iy$. Then $A + iB = \tan(x + iy)$, and therefore we get $A - iB = \tan(x - iy)$.

Now consider $2x = (x + iy) + (x - iy)$. Using $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$,

$$\begin{aligned} \tan 2x &= \tan[(x + iy) + (x - iy)] = \frac{\tan(x + iy) + \tan(x - iy)}{1 - \tan(x + iy) \tan(x - iy)} \\ &= \frac{(A + iB) + (A - iB)}{1 - (A + iB)(A - iB)} \\ &= \frac{2A}{1 - (A^2 - (iB)^2)} = \frac{2A}{1 - (A^2 + B^2)} \\ \Rightarrow 2x &= \tan^{-1} \left[\frac{2A}{1 - A^2 - B^2} \right] \Rightarrow \boxed{\text{Real part} = x = \frac{1}{2} \tan^{-1} \left(\frac{2A}{1 - A^2 - B^2} \right)} \end{aligned}$$

Similarly, $2iy = (x + iy) - (x - iy)$. Using $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$,

$$\begin{aligned} \tan(2iy) &= \tan[(x + iy) - (x - iy)] = \frac{\tan(x + iy) - \tan(x - iy)}{1 + \tan(x + iy) \tan(x - iy)} \\ \Rightarrow i \tanh 2y &= \frac{(A + iB) - (A - iB)}{1 + (A + iB)(A - iB)} = \frac{2iB}{1 + (A^2 - (iB)^2)} = \frac{i 2B}{1 + A^2 + B^2} \\ \Rightarrow \tanh 2y &= \frac{2B}{1 + A^2 + B^2} \Rightarrow 2y = \tanh^{-1} \left(\frac{2B}{1 + A^2 + B^2} \right) \\ \Rightarrow \boxed{\text{Imaginary part} = y = \frac{1}{2} \tanh^{-1} \left(\frac{2B}{1 + A^2 + B^2} \right)} \end{aligned}$$

■

Problem 4.2

If $x + iy = \tan(A + iB)$, prove that (i) $x^2 + y^2 + 2x \cot 2A = 1$ (ii) $x^2 + y^2 + 1 - 2y \coth 2B = 0$.

Solution. Given $x + iy = \tan(A + iB) \Rightarrow x - iy = \tan(A - iB)$.

(i) $2A = (A + iB) + (A - iB)$, so

$$\begin{aligned} \tan 2A &= \tan[(A + iB) + (A - iB)] = \frac{\tan(A + iB) + \tan(A - iB)}{1 - \tan(A + iB) \tan(A - iB)} \\ &= \frac{(x + iy) + (x - iy)}{1 - (x + iy)(x - iy)} \\ &= \frac{2x}{1 - (x^2 - (iy)^2)} = \frac{2x}{1 - x^2 - y^2} \\ \Rightarrow 1 - x^2 - y^2 &= \frac{2x}{\tan 2A} = 2x \cot 2A \Rightarrow \boxed{x^2 + y^2 + 2x \cot 2A = 1} \end{aligned}$$

(ii) Now $2iB = (A + iB) - (A - iB)$, so

$$\begin{aligned} \tan(2iB) &= \tan[(A + iB) - (A - iB)] = \frac{\tan(A + iB) - \tan(A - iB)}{1 + \tan(A + iB) \tan(A - iB)} \\ \Rightarrow i \tanh 2B &= \frac{(x + iy) - (x - iy)}{1 + (x + iy)(x - iy)} = \frac{2iy}{1 + (x^2 + y^2)} \\ \Rightarrow \tanh 2B &= \frac{2y}{1 + x^2 + y^2} \Rightarrow 1 + x^2 + y^2 = \frac{2y}{\tanh 2B} = 2y \coth 2B \end{aligned}$$

$$\Rightarrow \boxed{1 + x^2 + y^2 - 2y \coth 2B = 0}$$

■

Problem 4.3

If $u + iv = \cosh(x + iy)$, then prove that (i) $\frac{u^2}{\cosh^2 x} + \frac{v^2}{\sinh^2 x} = 1$ (ii) $\frac{u^2}{\cos^2 y} - \frac{v^2}{\sin^2 y} = 1$.

Solution. Given (using $\cosh x = \cos(ix)$)

$$\begin{aligned} u + iv &= \cosh(x + iy) = \cos i(x + iy) = \cos(ix - y) = \cos(ix) \cos y + \sin(ix) \sin y \\ &\Rightarrow u + iv = \cosh x \cos y + i \sinh x \sin y \quad (\because \sin(ix) = i \sinh x). \end{aligned}$$

Equating real and imaginary parts, we get

$$u = \cosh x \cos y \quad (1), \quad v = \sinh x \sin y \quad (2).$$

(i) From (1), $\frac{u}{\cosh x} = \cos y$, and from (2), $\frac{v}{\sinh x} = \sin y$. We know $\cos^2 y + \sin^2 y = 1$

$$\Rightarrow \boxed{\frac{u^2}{\cosh^2 x} + \frac{v^2}{\sinh^2 x} = 1}$$

(ii) Again from (1), $\cosh x = \frac{u}{\cos y}$, and from (2), $\sinh x = \frac{v}{\sin y}$. We know $\cosh^2 x - \sinh^2 x = 1$

$$\Rightarrow \boxed{\frac{u^2}{\cos^2 y} - \frac{v^2}{\sin^2 y} = 1}$$

■

Problem 4.4

If $x + iy = \sin(A + iB)$, then prove that (i) $\frac{x^2}{\sin^2 A} - \frac{y^2}{\cos^2 A} = 1$ (ii) $\frac{x^2}{\cosh^2 B} + \frac{y^2}{\sinh^2 B} = 1$.

Solution. $x + iy = \sin(A + iB) = \sin A \cos(iB) + \cos A \sin(iB) = \sin A \cosh B + i \cos A \sinh B$. Equating real and imaginary parts,

$$x = \sin A \cosh B \quad (1), \quad y = \cos A \sinh B \quad (2).$$

(i) From (1) and (2), $\cosh B = \frac{x}{\sin A}$, $\sinh B = \frac{y}{\cos A}$. Since $\cosh^2 B - \sinh^2 B = 1$,

$$\frac{x^2}{\sin^2 A} - \frac{y^2}{\cos^2 A} = 1.$$

(ii) From (1) and (2), $\sin A = \frac{x}{\cosh B}$, $\cos A = \frac{y}{\sinh B}$. Since $\sin^2 A + \cos^2 A = 1$,

$$\frac{x^2}{\cosh^2 B} + \frac{y^2}{\sinh^2 B} = 1.$$

■

Problem 4.5

If $\tan(x + iy) = u + iv$, then prove that $\frac{u}{v} = \frac{\sin 2x}{\sinh 2y}$.

Solution. Multiplying numerator and denominator by $2 \cos(x - iy)$,

$$u + iv = \tan(x + iy) = \frac{2 \sin(x + iy) \cos(x - iy)}{2 \cos(x + iy) \cos(x - iy)}$$

$$\begin{aligned}
&= \frac{\sin[(x+iy) + (x-iy)] + \sin[(x+iy) - (x-iy)]}{\cos[(x+iy) + (x-iy)] + \cos[(x+iy) - (x-iy)]} \\
&= \frac{\sin 2x + \sin(2iy)}{\cos 2x + \cos(2iy)} = \frac{\sin 2x + i \sinh 2y}{\cos 2x + \cosh 2y}
\end{aligned}$$

(using $2 \sin A \cos B = \sin(A+B) + \sin(A-B)$ and $2 \cos A \cos B = \cos(A+B) + \cos(A-B)$). Equating real and imaginary parts,

$$u = \frac{\sin 2x}{\cos 2x + \cosh 2y}, \quad v = \frac{\sinh 2y}{\cos 2x + \cosh 2y} \Rightarrow \frac{u}{v} = \frac{\sin 2x}{\sinh 2y}.$$

■

Problem 4.6

If $\cos(x+iy) = \cos \theta + i \sin \theta$, then prove that $\cos 2x + \cosh 2y = 2$.

Solution. Given

$$\begin{aligned}
\cos \theta + i \sin \theta &= \cos(x+iy) = \cos x \cos(iy) - \sin x \sin(iy) \\
\cos \theta + i \sin \theta &= \cos x \cosh y - i \sin x \sinh y.
\end{aligned}$$

Equating real and imaginary parts,

$$\cos \theta = \cos x \cosh y \quad (1), \quad \sin \theta = -\sin x \sinh y \quad (2).$$

We know $\cos^2 \theta + \sin^2 \theta = 1$

$$\begin{aligned}
&\Rightarrow \cos^2 x \cosh^2 y + \sin^2 x \sinh^2 y = 1 \\
&\Rightarrow \cos^2 x \cosh^2 y + (1 - \cos^2 x) \sinh^2 y = 1 \\
&\Rightarrow \cos^2 x (\cosh^2 y - \sinh^2 y) + \sinh^2 y = 1 \\
&\Rightarrow \cos^2 x + \sinh^2 y = 1 \quad (\because \cosh^2 y - \sinh^2 y = 1) \\
&\Rightarrow \frac{1 + \cos 2x}{2} + \frac{\cosh 2y - 1}{2} = 1 \quad \left(\because \cos^2 \theta = \frac{1 + \cos 2\theta}{2}, \sinh^2 y = \frac{\cosh 2y - 1}{2} \right) \\
&\Rightarrow 1 + \cos 2x + \cosh 2y - 1 = 2 \\
&\Rightarrow \boxed{\cos 2x + \cosh 2y = 2}
\end{aligned}$$

■

Problem 4.7

If $\sin(\theta+i\phi) = \tan \alpha + i \sec \alpha$, then prove that $\cos 2\theta \cosh 2\phi = 3$.

Solution. Given

$$\begin{aligned}
\tan \alpha + i \sec \alpha &= \sin(\theta+i\phi) = \sin \theta \cos(i\phi) + \cos \theta \sin(i\phi) \\
\tan \alpha + i \sec \alpha &= \sin \theta \cosh \phi + i \cos \theta \sinh \phi.
\end{aligned}$$

Equating real and imaginary parts, $\tan \alpha = \sin \theta \cosh \phi$ and $\sec \alpha = \cos \theta \sinh \phi$. We know $\sec^2 \alpha - \tan^2 \alpha = 1$

$$\begin{aligned}
&\Rightarrow \cos^2 \theta \sinh^2 \phi - \sin^2 \theta \cosh^2 \phi = 1 \\
&\Rightarrow \cos^2 \theta \sinh^2 \phi - (1 - \cos^2 \theta) \cosh^2 \phi = 1 \\
&\Rightarrow \cos^2 \theta \sinh^2 \phi - \cosh^2 \phi + \cos^2 \theta \cosh^2 \phi = 1 \\
&\Rightarrow \cos^2 \theta (\cosh^2 \phi + \sinh^2 \phi) - \cosh^2 \phi = 1 \\
&\Rightarrow \cos^2 \theta \cosh 2\phi - \cosh^2 \phi = 1 \\
&\Rightarrow \left(\frac{1 + \cos 2\theta}{2} \right) \cosh 2\phi - \left(\frac{1 + \cosh 2\phi}{2} \right) = 1 \quad \left(\because \cos^2 \theta = \frac{1 + \cos 2\theta}{2}, \cosh^2 \phi = \frac{1 + \cosh 2\phi}{2} \right) \\
&\Rightarrow \cosh 2\phi + \cos 2\theta \cosh 2\phi - 1 - \cosh 2\phi = 2 \\
&\Rightarrow \boxed{\cos 2\theta \cosh 2\phi = 3}
\end{aligned}$$

■

Problem 4.8

If $\cos(\alpha - i\beta) = \cos \alpha + i \sin \alpha$, then prove that $\sin \phi = \pm \sinh^2 \beta = \pm \sin^2 \alpha$.

Solution. Given

$$\begin{aligned}\cos \phi + i \sin \phi &= \cos(\alpha - i\beta) = \cos \alpha \cos(i\beta) + \sin \alpha \sin(i\beta) \\ \cos \phi + i \sin \phi &= \cos \alpha \cosh \beta + i \sin \alpha \sinh \beta.\end{aligned}$$

Equating, $\cos \phi = \cos \alpha \cosh \beta$ (1) and $\sin \phi = \sin \alpha \sinh \beta$ (2). We know $\cos^2 \phi + \sin^2 \phi = 1$

$$\begin{aligned}\Rightarrow \cos^2 \alpha \cosh^2 \beta + \sin^2 \alpha \sinh^2 \beta &= 1 \\ \Rightarrow (1 - \sin^2 \alpha) \cosh^2 \beta + \sin^2 \alpha \sinh^2 \beta &= 1 \\ \Rightarrow \cosh^2 \beta - \sin^2 \alpha \cosh^2 \beta + \sin^2 \alpha \sinh^2 \beta &= 1 \\ \Rightarrow \cosh^2 \beta - \sin^2 \alpha (\cosh^2 \beta - \sinh^2 \beta) &= 1 \\ \Rightarrow \cosh^2 \beta - \sin^2 \alpha &= 1 \\ \Rightarrow \sin^2 \alpha = \cosh^2 \beta - 1 = \sinh^2 \beta \quad (\because \cosh^2 \beta - \sinh^2 \beta = 1) \\ \Rightarrow \sin \alpha &= \pm \sinh \beta.\end{aligned} \tag{3}$$

Substituting this in (2),

$$\sin \phi = \sin \alpha \sinh \beta = \pm \sinh \beta \sinh \beta = \pm \sinh^2 \beta = \pm \sin^2 \alpha \quad (\because \text{by (3)}).$$

$$\Rightarrow \boxed{\sin \phi = \pm \sinh^2 \beta = \pm \sin^2 \alpha}$$

■

5 Inverse hyperbolic functions

Result 5.1 (Inverse hyperbolic functions)

- (i) $\sinh^{-1} x = \log_e (x + \sqrt{x^2 + 1})$
- (ii) $\cosh^{-1} x = \pm \log_e (x + \sqrt{x^2 - 1})$
- (iii) $\tanh^{-1} x = \frac{1}{2} \log_e \left(\frac{1+x}{1-x} \right)$

Recall: if $e^x = a$ then $\log_e e^x = \log_e a \Rightarrow x \log_e e = \log_e a \Rightarrow x = \log_e a$.

Proof. (i) Let $y = \sinh^{-1} x \Rightarrow \sinh y = x$, i.e.

$$\begin{aligned}x &= \frac{e^y - e^{-y}}{2} \Rightarrow 2x = e^y - \frac{1}{e^y} \Rightarrow 2x = \frac{e^{2y} - 1}{e^y} \\ \Rightarrow 2xe^y &= e^{2y} - 1 \Rightarrow e^{2y} - 2xe^y - 1 = 0,\end{aligned}$$

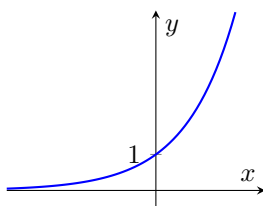
i.e. $(e^y)^2 - 2x(e^y) - 1 = 0$, which is a quadratic equation in e^y . Hence

$$e^y = \frac{2x \pm \sqrt{4x^2 - 4(1)(-1)}}{2(1)} = \frac{2x \pm 2\sqrt{x^2 + 1}}{2} = x \pm \sqrt{x^2 + 1},$$

i.e. $e^y = x + \sqrt{x^2 + 1}$ or $e^y = x - \sqrt{x^2 + 1}$. We know e^y is always positive, and $x - \sqrt{x^2 + 1} < 0$ (e.g. when $x = 3$: $3 + \sqrt{10} = 3 + 3.16 \dots > 0$ but $3 - \sqrt{10} = 3 - 3.16 \dots < 0$). Therefore we must have $e^y = x + \sqrt{x^2 + 1}$

$$\Rightarrow y = \log_e (x + \sqrt{x^2 + 1}) \Rightarrow \boxed{\sinh^{-1} x = \log_e (x + \sqrt{x^2 + 1})}$$

$$y = e^x > 0 \text{ for all } x$$



(ii) Let $y = \cosh^{-1} x \Rightarrow \cosh y = x$, i.e.

$$\begin{aligned} x &= \frac{e^y + e^{-y}}{2} \Rightarrow 2x = e^y + \frac{1}{e^y} = \frac{e^{2y} + 1}{e^y} \\ \Rightarrow 2xe^y &= e^{2y} + 1 \Rightarrow e^{2y} - 2xe^y + 1 = 0, \end{aligned}$$

which is a quadratic equation in e^y . So

$$e^y = \frac{2x \pm \sqrt{4x^2 - 4}}{2} = x \pm \sqrt{x^2 - 1} \Rightarrow e^y = x + \sqrt{x^2 - 1} \quad \text{or} \quad e^y = x - \sqrt{x^2 - 1}.$$

Here both values are positive for $x \geq 1$ (e.g. $3 + \sqrt{8} = 3 + 2.8\dots$, $3 - \sqrt{8} = 3 - 2.8\dots$). Moreover

$$x - \sqrt{x^2 - 1} = \frac{(x - \sqrt{x^2 - 1})(x + \sqrt{x^2 - 1})}{x + \sqrt{x^2 - 1}} = \frac{1}{x + \sqrt{x^2 - 1}},$$

so $y = \log_e (x + \sqrt{x^2 - 1})$ or $y = -\log_e (x + \sqrt{x^2 - 1})$. Hence

$$\boxed{\cosh^{-1} x = \pm \log_e (x + \sqrt{x^2 - 1})}$$

(iii) Let $y = \tanh^{-1} x \Rightarrow \tanh y = x$, i.e.

$$\begin{aligned} x &= \frac{e^y - e^{-y}}{e^y + e^{-y}} = \frac{e^{2y} - 1}{e^{2y} + 1} \Rightarrow \frac{1+x}{1-x} = \frac{(e^{2y} + 1) + (e^{2y} - 1)}{(e^{2y} + 1) - (e^{2y} - 1)} \quad (\text{componendo-dividendo}) \\ \Rightarrow \frac{1+x}{1-x} &= \frac{2e^{2y}}{2} = e^{2y} \Rightarrow 2y = \log_e \left(\frac{1+x}{1-x} \right) \Rightarrow \boxed{\tanh^{-1} x = \frac{1}{2} \log_e \left(\frac{1+x}{1-x} \right)} \end{aligned}$$

Note

In the same way one can show that $\coth^{-1} x = \frac{1}{2} \log_e \left(\frac{x+1}{x-1} \right)$ (Exercise).

5.1 Problems on inverse hyperbolic functions

Problem 5.1

If $\cos(x + iy) = r(\cos \alpha + i \sin \alpha)$, then prove that $y = \frac{1}{2} \log_e \left(\frac{\sin(x - \alpha)}{\sin(x + \alpha)} \right)$.

Solution. Given

$$\begin{aligned} r(\cos \alpha + i \sin \alpha) &= \cos(x + iy) = \cos x \cos(iy) - \sin x \sin(iy) \\ r \cos \alpha + ir \sin \alpha &= \cos x \cosh y - i \sin x \sinh y. \end{aligned}$$

Equating real and imaginary parts, we get

$$r \cos \alpha = \cos x \cosh y \quad (1), \quad r \sin \alpha = -\sin x \sinh y \quad (2).$$

$\frac{(2)}{(1)}$:

$$\begin{aligned}
 \frac{r \sin \alpha}{r \cos \alpha} &= \frac{-\sin x \sinh y}{\cos x \cosh y} \Rightarrow \tan \alpha = -\tan x \tanh y \Rightarrow \tanh y = -\frac{\tan \alpha}{\tan x} \\
 \Rightarrow y &= \tanh^{-1} \left(-\frac{\tan \alpha}{\tan x} \right) \\
 &= \frac{1}{2} \log_e \left[\frac{1 + \left(-\frac{\tan \alpha}{\tan x} \right)}{1 - \left(-\frac{\tan \alpha}{\tan x} \right)} \right] \quad \left(\because \tanh^{-1} x = \frac{1}{2} \log_e \frac{1+x}{1-x} \right) \\
 &= \frac{1}{2} \log_e \left[\frac{\tan x - \tan \alpha}{\tan x + \tan \alpha} \right] \\
 &= \frac{1}{2} \log_e \left[\frac{\frac{\sin x}{\cos x} - \frac{\sin \alpha}{\cos \alpha}}{\frac{\sin x}{\cos x} + \frac{\sin \alpha}{\cos \alpha}} \right] \\
 &= \frac{1}{2} \log_e \left[\frac{\sin x \cos \alpha - \cos x \sin \alpha}{\sin x \cos \alpha + \cos x \sin \alpha} \right] \\
 \Rightarrow y &= \frac{1}{2} \log_e \left[\frac{\sin(x - \alpha)}{\sin(x + \alpha)} \right].
 \end{aligned}$$

■

Problem 5.2

Prove that $u = \log_e \tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right)$ if and only if $\cosh u = \sec \theta$.

Solution. This proof contains an “if and only if” part; we can prove it in two parts.

Part I. Suppose $\cosh u = \sec \theta$. Then $u = \cosh^{-1}(\sec \theta)$. Since $\cosh^{-1} x = \pm \log_e (x + \sqrt{x^2 - 1})$ (taking the positive value),

$$\begin{aligned}
 u &= \log_e \left(\sec \theta + \sqrt{\sec^2 \theta - 1} \right) = \log_e (\sec \theta + \tan \theta) \\
 &= \log_e \left(\frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta} \right) = \log_e \left(\frac{1 + \sin \theta}{\cos \theta} \right).
 \end{aligned}$$

Using $\sin 2\theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}$ and $\cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$ (with $\theta \rightarrow \theta/2$),

$$\begin{aligned}
 u &= \log_e \left(\frac{1 + \frac{2 \tan \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}}}{\frac{1 - \tan^2 \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}}} \right) \\
 &= \log_e \left[\frac{1 + \tan^2 \frac{\theta}{2} + 2 \tan \frac{\theta}{2}}{1 - \tan^2 \frac{\theta}{2}} \right] \\
 &= \log_e \left[\frac{(1 + \tan \frac{\theta}{2})^2}{(1 + \tan \frac{\theta}{2})(1 - \tan \frac{\theta}{2})} \right] \\
 \Rightarrow u &= \log_e \left(\frac{1 + \tan \frac{\theta}{2}}{1 - \tan \frac{\theta}{2}} \right) \\
 &= \log_e \left(\frac{\tan \frac{\pi}{4} + \tan \frac{\theta}{2}}{1 - \tan \frac{\pi}{4} \tan \frac{\theta}{2}} \right) \\
 &= \log_e \left(\tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right) \right). \\
 \Rightarrow \boxed{u &= \log_e \tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right)}
 \end{aligned}$$

Part II. Suppose $u = \log_e \tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right)$. Then

$$\begin{aligned}
 e^u &= \tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right) = \frac{\tan \frac{\pi}{4} + \tan \frac{\theta}{2}}{1 - \tan \frac{\pi}{4} \tan \frac{\theta}{2}} \Rightarrow e^u = \frac{1 + \tan \frac{\theta}{2}}{1 - \tan \frac{\theta}{2}} \Rightarrow e^{-u} = \frac{1 - \tan \frac{\theta}{2}}{1 + \tan \frac{\theta}{2}}. \\
 \Rightarrow \cosh u &= \frac{1}{2} [e^u + e^{-u}] = \frac{1}{2} \left[\frac{1 + \tan \frac{\theta}{2}}{1 - \tan \frac{\theta}{2}} + \frac{1 - \tan \frac{\theta}{2}}{1 + \tan \frac{\theta}{2}} \right] \\
 &= \frac{1}{2} \left[\frac{(1 + \tan \frac{\theta}{2})^2 + (1 - \tan \frac{\theta}{2})^2}{(1 - \tan \frac{\theta}{2})(1 + \tan \frac{\theta}{2})} \right] \\
 &= \frac{1}{2} \left[\frac{1 + \tan^2 \frac{\theta}{2} + 2 \tan \frac{\theta}{2} + 1 + \tan^2 \frac{\theta}{2} - 2 \tan \frac{\theta}{2}}{1 - \tan^2 \frac{\theta}{2}} \right] \\
 &= \frac{1}{2} \left[\frac{2(1 + \tan^2 \frac{\theta}{2})}{1 - \tan^2 \frac{\theta}{2}} \right] \\
 &= \frac{1 + \tan^2 \frac{\theta}{2}}{1 - \tan^2 \frac{\theta}{2}} = \frac{1}{\cos \theta} \Rightarrow \boxed{\cosh u = \sec \theta}
 \end{aligned}$$

■

Problem 5.3

Expand $\cosh^6 \theta$ (in terms of hyperbolic cosines of multiples of θ).

Solution. We know that $\cosh \theta = \frac{e^\theta + e^{-\theta}}{2}$ and $\cosh n\theta = \frac{e^{n\theta} + e^{-n\theta}}{2}$.

$$\begin{aligned}
 \Rightarrow \cosh^6 \theta &= \frac{1}{2^6} (e^\theta + e^{-\theta})^6 \\
 &= \frac{1}{2^6} [e^{6\theta} + {}^6C_1 e^{5\theta} e^{-\theta} + {}^6C_2 e^{4\theta} e^{-2\theta} + {}^6C_3 e^{3\theta} e^{-3\theta} + {}^6C_4 e^{2\theta} e^{-4\theta} + {}^6C_5 e^\theta e^{-5\theta} + {}^6C_6 e^{-6\theta}] \\
 &= \frac{1}{2^6} [(e^{6\theta} + e^{-6\theta}) + 6(e^{4\theta} + e^{-4\theta}) + 15(e^{2\theta} + e^{-2\theta}) + 20] \\
 &= \frac{1}{2^6} [2 \cosh 6\theta + 6(2 \cosh 4\theta) + 15(2 \cosh 2\theta) + 20] \\
 &= \frac{1}{2^5} [\cosh 6\theta + 6 \cosh 4\theta + 15 \cosh 2\theta + 10] \\
 \Rightarrow \boxed{2^5 \cosh^6 \theta = \cosh 6\theta + 6 \cosh 4\theta + 15 \cosh 2\theta + 10}
 \end{aligned}$$

■

Exercise. Similarly show that $\sinh^5 \theta = \frac{1}{2^4} [\sinh 5\theta - 5 \sinh 3\theta + 10 \sinh \theta]$.

Tutorial problem set – 2 (Hyperbolic functions)

- (1) Separate the real and imaginary parts of (i) $\operatorname{cosec}(x + iy)$ (ii) $\tanh(x + iy)$ (iii) $\cos^{-1}(\cos \theta + i \sin \theta)$.
- (2) If $\sin(\theta + i\phi) = \cos \alpha + i \sin \alpha$, then prove that $\cos^2 \theta = \pm \sin \alpha$.
- (3) If $x + iy = \cos(u + iv)$, prove that $(1 + x)^2 + y^2 = (\cosh v + \cos u)^2$.
- (4) If $A + iB = \tan^{-1}(x + iy)$, prove that $B = \frac{1}{4} \log_e \frac{x^2 + (1 + y)^2}{x^2 + (1 - y)^2}$.
- (5) If $\tan(\theta + i\phi) = \sin(x + iy)$, prove that $\sin 2\theta \coth y = \sinh 2\phi \cot x$.

(6) Prove that $\tanh \frac{x}{2} = \tan \frac{\theta}{2}$ if and only if $x = \log_e \tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right)$.

Hints. (1)(i) $\operatorname{cosec}(x+iy) = \frac{1}{\sin(x+iy)} \times \frac{\sin(x-iy)}{\sin(x-iy)} = \frac{\sin(x-iy)}{\frac{1}{2}[\cos(2iy) - \cos 2x]}$, using $\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$; then expand $\sin(x-iy)$.

(1)(ii) $\tanh(x+iy) = \frac{\sinh(x+iy)}{\cosh(x+iy)} = \frac{-i \sin i(x+iy)}{\cos i(x+iy)} = -i \frac{\sin(ix-y)}{\cos(ix-y)}$; multiply numerator and denominator by $2 \cos(ix+y)$.

(1)(iii) Let $\cos^{-1}(\cos \theta + i \sin \theta) = x + iy$. Then $\cos \theta + i \sin \theta = \cos(x+iy) = \cos x \cosh y - i \sin x \sinh y$, so $\cos \theta = \cos x \cosh y$ (1), $\sin \theta = -\sin x \sinh y$ (2). Use $\cos^2 \theta + \sin^2 \theta = 1$ to get $\sin^2 x = \cos \theta$ (i.e. $\cos^2 x + \sinh^2 y = 1$), and $\frac{(2)}{(1)}$ gives $\tan \theta = -\tan x \tanh y$, i.e. $y = \tanh^{-1} \left(-\frac{\tan \theta}{\tan x} \right)$.

6 Logarithm of a complex number

We know that any complex number $x + iy$ can be written in polar form as

$$x + iy = r(\cos \theta + i \sin \theta), \quad \text{where } r = \sqrt{x^2 + y^2} \text{ and } \theta = \tan^{-1} \left(\frac{y}{x} \right),$$

$$\boxed{x + iy = re^{i\theta}} \quad (1)$$

$$\Rightarrow \log_e(x + iy) = \log_e(re^{i\theta}) = \log_e r + \log_e e^{i\theta} = \log_e r + i\theta$$

$$\Rightarrow \log_e(x + iy) = \log_e \sqrt{x^2 + y^2} + i \tan^{-1} \left(\frac{y}{x} \right)$$

$$\boxed{\log_e(x + iy) = \frac{1}{2} \log_e(x^2 + y^2) + i \tan^{-1} \left(\frac{y}{x} \right)} \quad (2)$$

Now, we know that $e^{i\pi} = \cos \pi + i \sin \pi = -1 + i(0) = -1$, and

$$e^{i2\pi} = \cos 2\pi + i \sin 2\pi = 1 + i(0) = 1 \Rightarrow e^{i2n\pi} = 1, \quad n = 0, 1, 2, 3, \dots$$

From (1) we see that

$$x + iy = re^{i\theta} = re^{i\theta} e^{i2n\pi} \quad (\because e^{i2n\pi} = 1) \Rightarrow x + iy = re^{i(\theta+2n\pi)}$$

$$\begin{aligned} \Rightarrow \operatorname{Log}_e(x + iy) &= \log_e r + \log_e e^{i(\theta+2n\pi)} = \log_e r + i(\theta + 2n\pi) \\ &= \frac{1}{2} \log_e(x^2 + y^2) + i(\theta + 2n\pi), \quad n = 0, 1, 2, 3, \dots \end{aligned}$$

$$\Rightarrow \boxed{\operatorname{Log}_e(x + iy) = \frac{1}{2} \log_e(x^2 + y^2) + i \left(\tan^{-1} \left(\frac{y}{x} \right) + 2n\pi \right)}, \quad n = 0, 1, 2, 3, \dots \quad (3)$$

When $n = 0$, $\operatorname{Log}_e(x + iy) = \frac{1}{2} \log_e(x^2 + y^2) + i \tan^{-1} \left(\frac{y}{x} \right) = \log_e(x + iy)$ (by (2)). From equation (3) we can understand that the logarithm of a complex number is a *many-valued* function.

Definition 6.1 (Principal and general values)

Equation (2) is called the **principal value** of the logarithm of a complex number:

$$\log_e(x + iy) = \frac{1}{2} \log_e(x^2 + y^2) + i \tan^{-1} \left(\frac{y}{x} \right) = \log_e r + i\theta.$$

Equation (3) is called the **general value** of the logarithm of a complex number:

$$\operatorname{Log}_e(x + iy) = \frac{1}{2} \log_e(x^2 + y^2) + i \left(\tan^{-1} \left(\frac{y}{x} \right) + 2n\pi \right) = \log_e r + i(\theta + 2n\pi),$$

where $r = \sqrt{x^2 + y^2}$, $\theta = \tan^{-1} \left(\frac{y}{x} \right)$ and $n = 0, 1, 2, 3, \dots$

6.1 Problems

Problem 6.1

Determine the general logarithm of the following complex numbers: (1) $2 + 3i$ (2) i (3) $\frac{1+i}{1-i}$ (4) i^k , k is an integer.

Solution. (1) Given $2+3i = x+iy$. Then $\text{Log}_e(2+3i) = \log_e r + i(\theta + 2n\pi)$, where $r = \sqrt{x^2 + y^2} = \sqrt{4+9} = \sqrt{13}$ and $\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{3}{2}\right)$.

$$\Rightarrow \boxed{\text{Log}_e(2+3i) = \log_e \sqrt{13} + i\left(\tan^{-1}\left(\frac{3}{2}\right) + 2n\pi\right)}$$

(2) Given $i = 0 + i(1) = x + iy$. $\text{Log}_e(i) = \log_e r + i(\theta + 2n\pi)$ where $r = \sqrt{x^2 + y^2} = \sqrt{0+1} = 1$, $\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{1}{0}\right) = \tan^{-1}(\infty) = \tan^{-1}\left(\tan\frac{\pi}{2}\right) \Rightarrow \theta = \frac{\pi}{2}$.

$$\Rightarrow \text{Log}_e i = \log_e 1 + i\left(\frac{\pi}{2} + 2n\pi\right) = i\left(\frac{\pi}{2} + 2n\pi\right) \quad (\because \log 1 = 0) \Rightarrow \boxed{\text{Log}_e i = i(4n+1)\frac{\pi}{2}}, \quad n = 0, 1, 2, \dots$$

(3) Given

$$\frac{1+i}{1-i} = \frac{1+i}{1-i} \times \frac{1+i}{1+i} = \frac{1-1+2i}{1-(-1)} = \frac{2i}{2} = i \Rightarrow \text{Log}_e\left(\frac{1+i}{1-i}\right) = \text{Log}_e(i) = i(4n+1)\frac{\pi}{2} \quad (\because \text{by (2)}).$$

(4)

$$\begin{aligned} i^k &= (0 + i(1))^k = \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)^k = \cos \frac{k\pi}{2} + i \sin \frac{k\pi}{2} \quad (\because \text{by De Moivre's rule}) \\ &= e^{ik\pi/2} = 1 \cdot e^{ik\pi/2} = re^{i\theta}, \quad \text{where } r = 1, \theta = \frac{k\pi}{2}. \end{aligned}$$

$$\Rightarrow \text{Log } i^k = \log_e r + i(\theta + 2n\pi) = \log_e 1 + i\left(\frac{k\pi}{2} + 2n\pi\right) \Rightarrow \boxed{\text{Log}_e i^k = i\frac{\pi}{2}(4n+k)}$$

■

Problem 6.2

Separate the real and imaginary parts of $(\alpha + i\beta)^{x+iy}$.

Solution.

$$\begin{aligned} (\alpha + i\beta)^{x+iy} &= e^{\text{Log}(\alpha+i\beta)^{x+iy}} = e^{(x+iy)\text{Log}(\alpha+i\beta)} \\ &= e^{(x+iy)(\log r + i(\theta + 2n\pi))}, \quad r = \sqrt{\alpha^2 + \beta^2}, \theta = \tan^{-1}\left(\frac{\beta}{\alpha}\right) \\ &= e^{(x \log r - y(\theta + 2n\pi)) + i(y \log r + x(\theta + 2n\pi))} \\ &= e^{x \log r - y(\theta + 2n\pi)} e^{i(y \log r + x(\theta + 2n\pi))} \\ &= e^A e^{iB}, \quad \text{where } A = x \log_e r - y(\theta + 2n\pi), B = y \log_e r + x(\theta + 2n\pi) \\ &= e^A (\cos B + i \sin B) \quad (\because e^{i\theta} = \cos \theta + i \sin \theta) \\ &= e^A \cos B + i e^A \sin B. \\ \Rightarrow \text{Real part} &= e^A \cos B, \quad \text{Imaginary part} = e^A \sin B, \end{aligned}$$

where $A = x \log_e r - y(\theta + 2n\pi)$, $B = y \log_e r + x(\theta + 2n\pi)$, $r = \sqrt{\alpha^2 + \beta^2}$ and $\theta = \tan^{-1}\left(\frac{\beta}{\alpha}\right)$.

■

Note**Recall.**

General value: $\text{Log}(x + iy) = \frac{1}{2} \log_e(x^2 + y^2) + i\left(\tan^{-1}\left(\frac{y}{x}\right) + 2n\pi\right) = \log_e r + i(\theta + 2n\pi)$, $n = 0, 1, 2, \dots$

Principal value: $\log_e(x + iy) = \frac{1}{2} \log_e(x^2 + y^2) + i \tan^{-1}\left(\frac{y}{x}\right) = \log_e r + i\theta$, where $r = \sqrt{x^2 + y^2}$, $\theta = \tan^{-1}\left(\frac{y}{x}\right)$.

Problem 6.3

Prove that $i^i = e^{-(4n+1)\pi/2}$.

Solution. Let $i^i = x$. Then

$$\text{Log}_e i^i = \text{Log}_e x \Rightarrow \text{Log}_e x = i \text{Log}_e i = i \left[i \frac{\pi}{2} (4n+1) \right] \quad (\because \text{by Problem 1(2)})$$

$$\text{Log}_e x = -\frac{\pi}{2} (4n+1) \Rightarrow x = e^{-\frac{\pi}{2} (4n+1)} \Rightarrow \boxed{i^i = e^{-\frac{\pi}{2} (4n+1)}}$$

■

Problem 6.4

If $i^{i^{i^{\dots\infty}}} = x + iy$, prove that (i) $\tan \frac{\pi x}{2} = \frac{y}{x}$ (ii) $x^2 + y^2 = e^{-\pi y}$.

Solution. Given $i^{i^{i^{\dots\infty}}} = x + iy \Rightarrow i^{x+iy} = x + iy$ (the exponent is again the same infinite tower). Taking (principal) logarithms,

$$\begin{aligned} \Rightarrow \log_e i^{x+iy} &= \log_e (x + iy) \Rightarrow \log_e (x + iy) = (x + iy) \log_e i \\ \Rightarrow \frac{1}{2} \log_e (x^2 + y^2) + i \tan^{-1} \left(\frac{y}{x} \right) &= (x + iy) \log_e (0 + i(1)) = (x + iy) \left(\frac{1}{2} \log(1) + i \tan^{-1} \left(\frac{1}{0} \right) \right) \\ &= (x + iy) \left(i \frac{\pi}{2} \right) \quad (\because \log 1 = 0) \\ &= i \frac{\pi x}{2} - \frac{\pi y}{2}. \end{aligned}$$

Equating real and imaginary parts on both sides,

$$\frac{1}{2} \log_e (x^2 + y^2) = -\frac{\pi y}{2} \quad \text{and} \quad \tan^{-1} \left(\frac{y}{x} \right) = \frac{\pi x}{2} \Rightarrow \boxed{x^2 + y^2 = e^{-\pi y}} \quad \text{and} \quad \boxed{\frac{y}{x} = \tan \frac{\pi x}{2}}$$

■

Problem 6.5

Prove that the real part of the principal value of $i^{\log(1+i)}$ is equal to $e^{-\pi^2/8} \cos \left(\frac{\pi}{4} \log_e 2 \right)$.

Solution. Let $i^{\log(1+i)} = x + iy$. Then

$$\begin{aligned} \log_e (x + iy) &= \log_e i^{\log(1+i)} = \log_e (1 + i) \times \log_e i \\ &= \left[\frac{1}{2} \log(2) + i \tan^{-1} \left(\frac{1}{1} \right) \right] \times \left(i \frac{\pi}{2} \right) \quad (\because \text{by Problem 1(2), } n = 0) \\ &= \left(\frac{1}{2} \log_e 2 + i \frac{\pi}{4} \right) \left(i \frac{\pi}{2} \right) = i \frac{\pi}{4} \log_e 2 - \frac{\pi^2}{8} \\ \Rightarrow x + iy &= e^{\left(i \frac{\pi}{4} \log_e 2 - \frac{\pi^2}{8} \right)} = e^{i \frac{\pi}{4} \log_e 2} e^{-\pi^2/8} \\ &= e^{-\pi^2/8} \left(\cos \left(\frac{\pi}{4} \log_e 2 \right) + i \sin \left(\frac{\pi}{4} \log_e 2 \right) \right) \quad (\because e^{i\theta} = \cos \theta + i \sin \theta) \\ \Rightarrow \boxed{\text{Real part} = x = e^{-\pi^2/8} \cos \left(\frac{\pi}{4} \log_e 2 \right)} \end{aligned}$$

■

Problem 6.6

If $\tan \log_e (x + iy) = a + ib$, where $a^2 + b^2 \neq 1$, prove that $\tan \log_e (x^2 + y^2) = \frac{2a}{1 - a^2 - b^2}$.

Solution. Given $\tan(\log_e(x + iy)) = a + ib$

$$\Rightarrow \tan\left(\frac{1}{2}\log_e(x^2 + y^2) + i\tan^{-1}\left(\frac{y}{x}\right)\right) = a + ib$$

$$\Rightarrow \tan\left(\frac{1}{2}\log_e(x^2 + y^2) - i\tan^{-1}\left(\frac{y}{x}\right)\right) = a - ib,$$

i.e. $\tan(\theta + i\phi) = a + ib$ and $\tan(\theta - i\phi) = a - ib$, where $\theta = \frac{1}{2}\log_e(x^2 + y^2)$, $\phi = \tan^{-1}\left(\frac{y}{x}\right)$.

$$\therefore 2\theta = (\theta + i\phi) + (\theta - i\phi)$$

$$\Rightarrow \tan(2\theta) = \tan[(\theta + i\phi) + (\theta - i\phi)] = \frac{\tan(\theta + i\phi) + \tan(\theta - i\phi)}{1 - \tan(\theta + i\phi)\tan(\theta - i\phi)}$$

$$= \frac{(a + ib) + (a - ib)}{1 - (a + ib)(a - ib)}$$

$$\tan 2\theta = \frac{2a}{1 - (a^2 + b^2)} \Rightarrow \tan\left(2 \cdot \frac{1}{2}\log_e(x^2 + y^2)\right) = \frac{2a}{1 - (a^2 + b^2)} \quad (\because \theta = \frac{1}{2}\log_e(x^2 + y^2))$$

$$\Rightarrow \boxed{\tan(\log_e(x^2 + y^2)) = \frac{2a}{1 - (a^2 + b^2)}}, \quad \text{where } a^2 + b^2 \neq 1.$$

■

Problem 6.7

Prove that $\tan\left[i\log_e\left(\frac{a - ib}{a + ib}\right)\right] = \frac{2ab}{a^2 - b^2}$.

Solution. Now

$$\begin{aligned} \log_e\left(\frac{a - ib}{a + ib}\right) &= \log(a - ib) - \log(a + ib) \\ &= \left[\frac{1}{2}\log(a^2 + b^2) + i\tan^{-1}\left(\frac{-b}{a}\right)\right] - \left[\frac{1}{2}\log(a^2 + b^2) + i\tan^{-1}\left(\frac{b}{a}\right)\right] \\ &= \frac{1}{2}\log(a^2 + b^2) - i\tan^{-1}\left(\frac{b}{a}\right) - \frac{1}{2}\log(a^2 + b^2) - i\tan^{-1}\left(\frac{b}{a}\right) \\ \Rightarrow \log_e\left(\frac{a - ib}{a + ib}\right) &= -2i\tan^{-1}\left(\frac{b}{a}\right). \end{aligned}$$

$$\begin{aligned} \Rightarrow \tan\left[i\log_e\left(\frac{a - ib}{a + ib}\right)\right] &= \tan\left(-2i^2\tan^{-1}\left(\frac{b}{a}\right)\right) = \tan\left(2\tan^{-1}\left(\frac{b}{a}\right)\right) \\ &= \tan(2\theta), \quad \text{where } \theta = \tan^{-1}\left(\frac{b}{a}\right) \Rightarrow \frac{b}{a} = \tan \theta \\ &= \frac{2\tan \theta}{1 - \tan^2 \theta} = \frac{2\frac{b}{a}}{1 - \frac{b^2}{a^2}} = \frac{\frac{2b}{a}}{\frac{a^2 - b^2}{a^2}} \end{aligned}$$

$$\Rightarrow \boxed{\tan\left[i\log_e\left(\frac{a - ib}{a + ib}\right)\right] = \frac{2ab}{a^2 - b^2}}$$

■

Problem 6.8

If $\alpha + i\beta = b^{x+iy}$, prove that $\frac{y}{x} = \frac{2\tan^{-1}\left(\frac{\beta}{\alpha}\right)}{\log_e(\alpha^2 + \beta^2)}$.

Solution. Taking (principal) logarithms, $\log_e(\alpha + i\beta) = (x + iy) \log_e b$, i.e.

$$\frac{1}{2} \log_e(\alpha^2 + \beta^2) + i \tan^{-1} \left(\frac{\beta}{\alpha} \right) = x \log_e b + iy \log_e b.$$

Equating real and imaginary parts, $\frac{1}{2} \log_e(\alpha^2 + \beta^2) = x \log_e b$ and $\tan^{-1} \left(\frac{\beta}{\alpha} \right) = y \log_e b$. Dividing,

$$\frac{y}{x} = \frac{\tan^{-1}(\beta/\alpha)}{\frac{1}{2} \log_e(\alpha^2 + \beta^2)} = \frac{2 \tan^{-1}(\beta/\alpha)}{\log_e(\alpha^2 + \beta^2)}.$$

■

Problem 6.9

If $\log_e \sin(x + iy) = a + ib$, prove that $\cosh 2y - \cos 2x = 2e^{2a}$.

Solution. $\log_e \sin(x + iy) = a + ib \Rightarrow \sin(x + iy) = e^{a+ib} = e^a e^{ib} = e^a (\cos b + i \sin b)$. Also $\sin(x + iy) = \sin x \cosh y + i \cos x \sinh y$. Equating,

$$\sin x \cosh y = e^a \cos b, \quad \cos x \sinh y = e^a \sin b.$$

Squaring and adding,

$$\begin{aligned} e^{2a} &= \sin^2 x \cosh^2 y + \cos^2 x \sinh^2 y = \sin^2 x (1 + \sinh^2 y) + (1 - \sin^2 x) \sinh^2 y = \sin^2 x + \sinh^2 y \\ &= \frac{1 - \cos 2x}{2} + \frac{\cosh 2y - 1}{2} = \frac{\cosh 2y - \cos 2x}{2} \Rightarrow \cosh 2y - \cos 2x = 2e^{2a}. \end{aligned}$$

■

Problem 6.10

Separate the real and imaginary parts of $\frac{(\sqrt{3} + i)^{\sqrt{3}-i}}{(\sqrt{3} - i)^{\sqrt{3}+i}}$ by taking the principal logarithm.

Solution. Let $\frac{(\sqrt{3} + i)^{\sqrt{3}-i}}{(\sqrt{3} - i)^{\sqrt{3}+i}} = x + iy$. Taking logarithms,

$$\log_e(x + iy) = (\sqrt{3} - i) \log(\sqrt{3} + i) - (\sqrt{3} + i) \log(\sqrt{3} - i).$$

Here $\log(\sqrt{3} \pm i) = \frac{1}{2} \log(3 + 1) \pm i \tan^{-1} \frac{1}{\sqrt{3}} = \log 2 \pm i \frac{\pi}{6}$. Hence

$$\begin{aligned} \log_e(x + iy) &= (\sqrt{3} - i) \left(\log 2 + i \frac{\pi}{6} \right) - (\sqrt{3} + i) \left(\log 2 - i \frac{\pi}{6} \right) \\ &= \left(\sqrt{3} \log 2 + i \frac{\sqrt{3}\pi}{6} - i \log 2 + \frac{\pi}{6} \right) - \left(\sqrt{3} \log 2 - i \frac{\sqrt{3}\pi}{6} + i \log 2 + \frac{\pi}{6} \right) \\ &= i \left(\frac{\sqrt{3}\pi}{3} - 2 \log 2 \right) = i \left(\frac{\pi}{\sqrt{3}} - \log 4 \right). \end{aligned}$$

$$\Rightarrow x + iy = e^{i \left(\frac{\pi}{\sqrt{3}} - \log 4 \right)} \Rightarrow x = \cos \left(\log 4 - \frac{\pi}{\sqrt{3}} \right), \quad y = \sin \left(\frac{\pi}{\sqrt{3}} - \log 4 \right) = -\sin \left(\log 4 - \frac{\pi}{\sqrt{3}} \right).$$

■

Problem 6.11

If $\log_e \cos(\alpha - i\beta) = x + iy$, prove that $x = \frac{1}{2} \log_e \left(\frac{\cos 2\alpha + \cosh 2\beta}{2} \right)$ and $\beta = \frac{1}{2} \log_e \left[\frac{\sin(\alpha + y)}{\sin(\alpha - y)} \right]$.

Solution. $\cos(\alpha - i\beta) = \cos \alpha \cosh \beta + i \sin \alpha \sinh \beta$. So

$$x + iy = \log(\cos \alpha \cosh \beta + i \sin \alpha \sinh \beta) = \frac{1}{2} \log(\cos^2 \alpha \cosh^2 \beta + \sin^2 \alpha \sinh^2 \beta) + i \tan^{-1} \left(\frac{\sin \alpha \sinh \beta}{\cos \alpha \cosh \beta} \right).$$

Now $\cos^2 \alpha \cosh^2 \beta + \sin^2 \alpha \sinh^2 \beta = \frac{(1+\cos 2\alpha)(\cosh 2\beta+1)+(1-\cos 2\alpha)(\cosh 2\beta-1)}{4} = \frac{\cos 2\alpha + \cosh 2\beta}{2}$, so

$$x = \frac{1}{2} \log_e \left(\frac{\cos 2\alpha + \cosh 2\beta}{2} \right).$$

Also $y = \tan^{-1}(\tan \alpha \tanh \beta) \Rightarrow \tan y = \tan \alpha \tanh \beta \Rightarrow \tanh \beta = \frac{\tan y}{\tan \alpha}$, so

$$\beta = \tanh^{-1} \left(\frac{\tan y}{\tan \alpha} \right) = \frac{1}{2} \log \left(\frac{1 + \frac{\tan y}{\tan \alpha}}{1 - \frac{\tan y}{\tan \alpha}} \right) = \frac{1}{2} \log \left(\frac{\tan \alpha + \tan y}{\tan \alpha - \tan y} \right) = \frac{1}{2} \log_e \left[\frac{\sin(\alpha + y)}{\sin(\alpha - y)} \right].$$

■

Exercise problems

- (1) If $\frac{(1+i)^{x+iy}}{(1-i)^{x-iy}} = A + iB$, then prove that $\tan^{-1} \left(\frac{B}{A} \right) = \frac{\pi x}{2} + y \log_e 2$.
- (2) Prove that $\text{Log}_e(1 + \cos 2\theta + i \sin 2\theta) = \log_e(2 \cos \theta) + i(2n\pi + \theta)$.
- (3) Find the general value of the logarithm of (i) $1 + \sqrt{3}i$ (ii) $1 - i$.